

Decision Trees

Introduction to Machine Learning

Referensi

- Machine Learning
— *Tom M. Mitchell*
- An Introduction to Statistical Learning
— *Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani*
- The Elements of Statistical Learning
— *Trevor Hastie, Robert Tibshirani, Jerome Friedman*
- Decision Tree
— *Eric Eaton, University of Pennsylvania*

Outcomes

- Siswa dapat **menjelaskan** cara kerja Decision Tree untuk menyelesaikan regresi & klasifikasi
- Siswa dapat **menjelaskan** kondisi overfit pada Decision Tree
- Siswa dapat **menjelaskan** cara mengatasi overfit pada Decision Tree
- Siswa dapat **menjelaskan** limitasi dari Decision Tree

Outline

- The Basic of Decision Tree
- Classification Trees
- Regression Trees
- Overfitting in Decision Tree
- Decision Tree Complexity
- Tree Pruning
- Limitation of Decision Tree

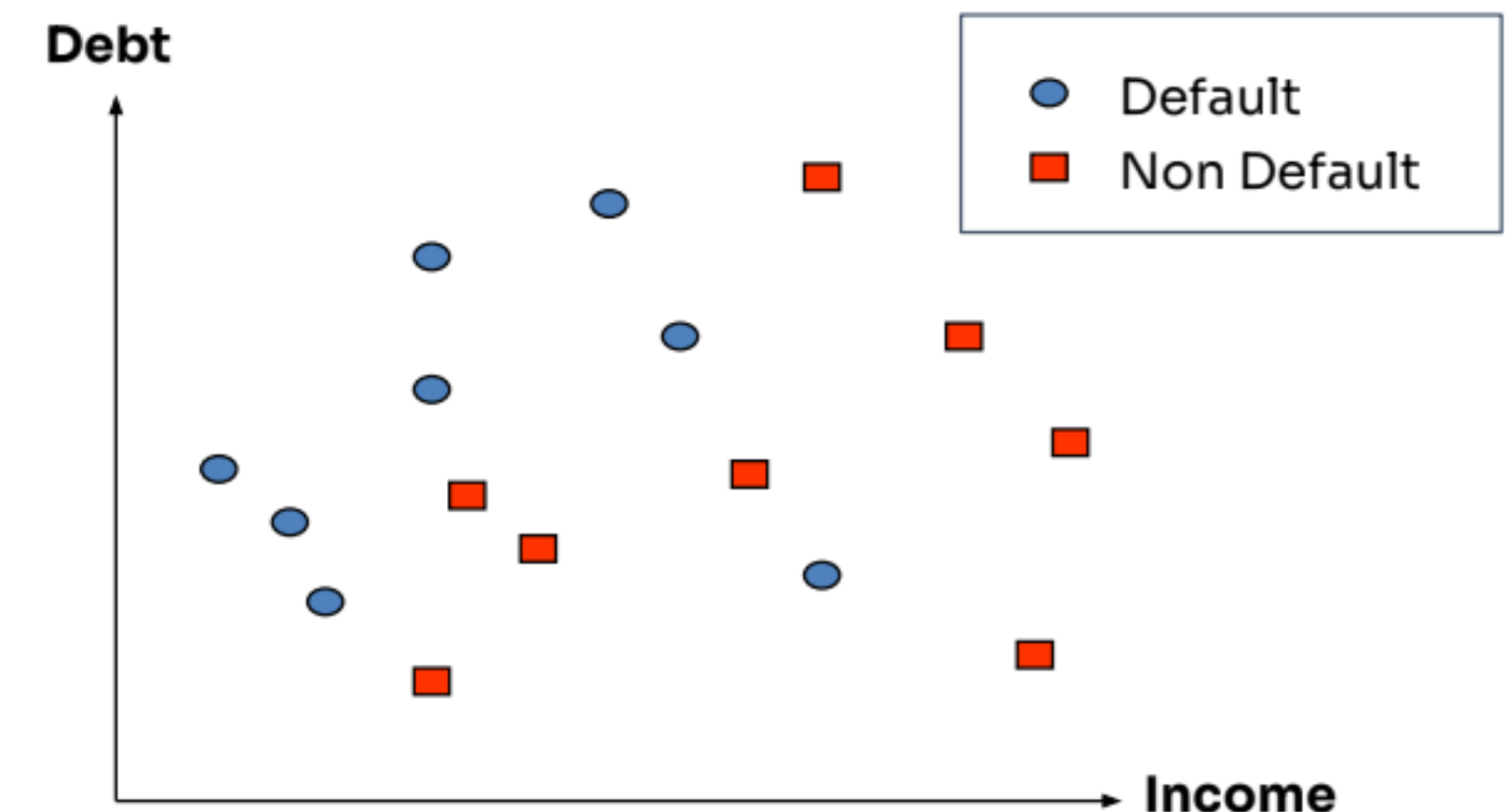
The Basic of Decision Tree

What is Decision Tree?

Case : Credit default
Task : Klasifikasi
Data

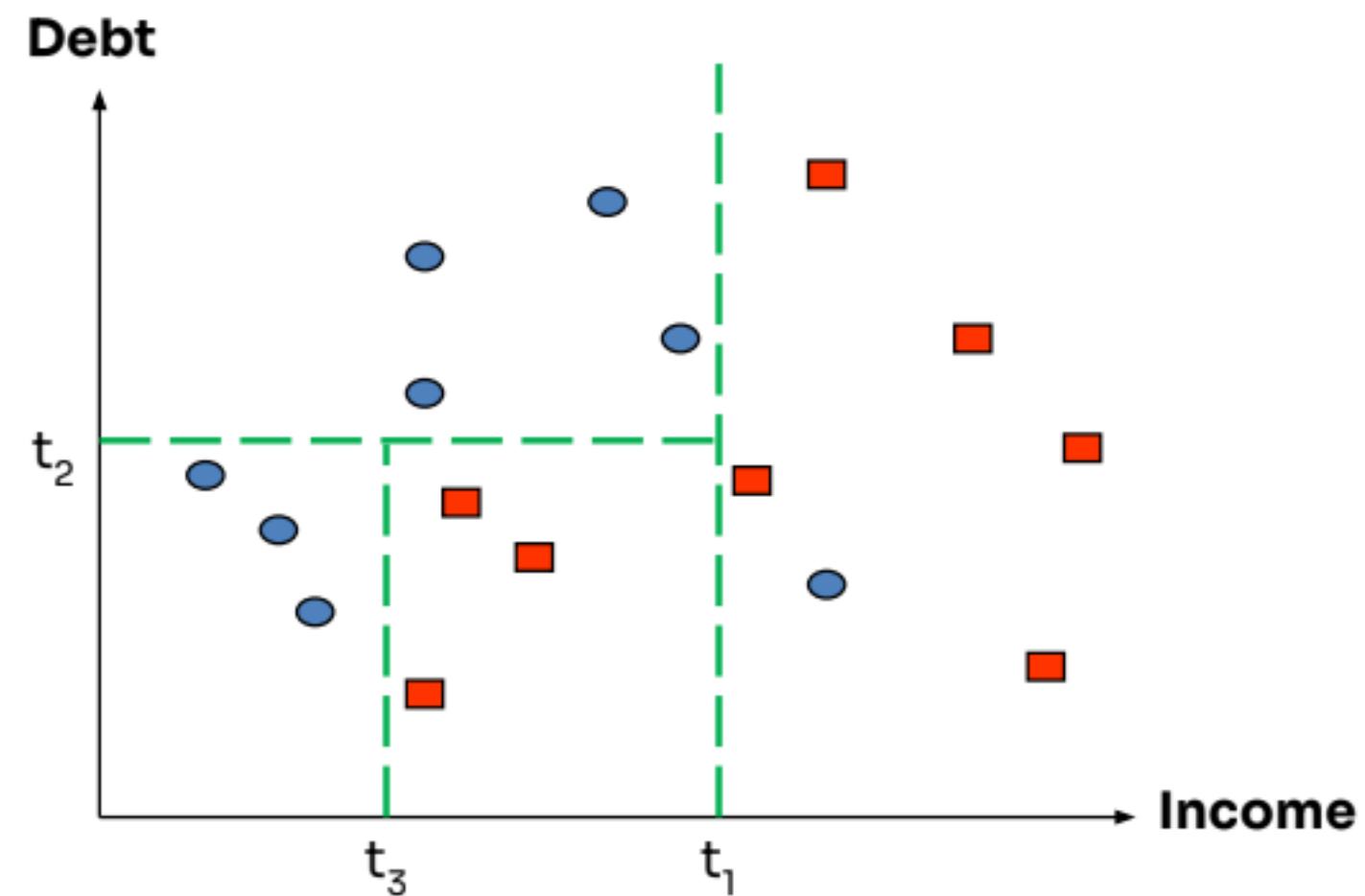
Age	23 years
Gender	Male
Annual salary (income)	\$ 30,000
Years in residence	1 year
Current debt	\$ 15,000
...	...
Status	Default

Applicant Data



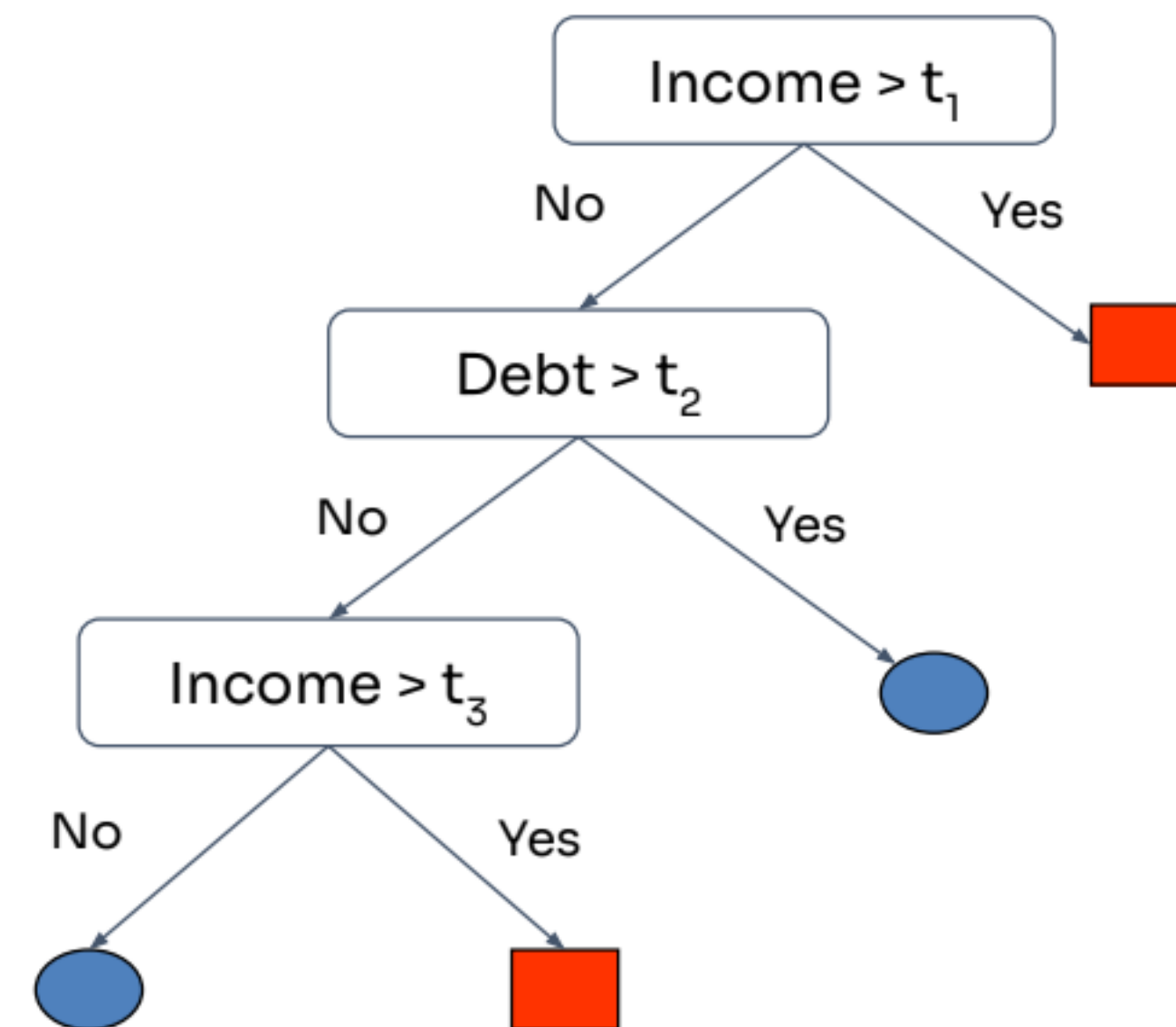
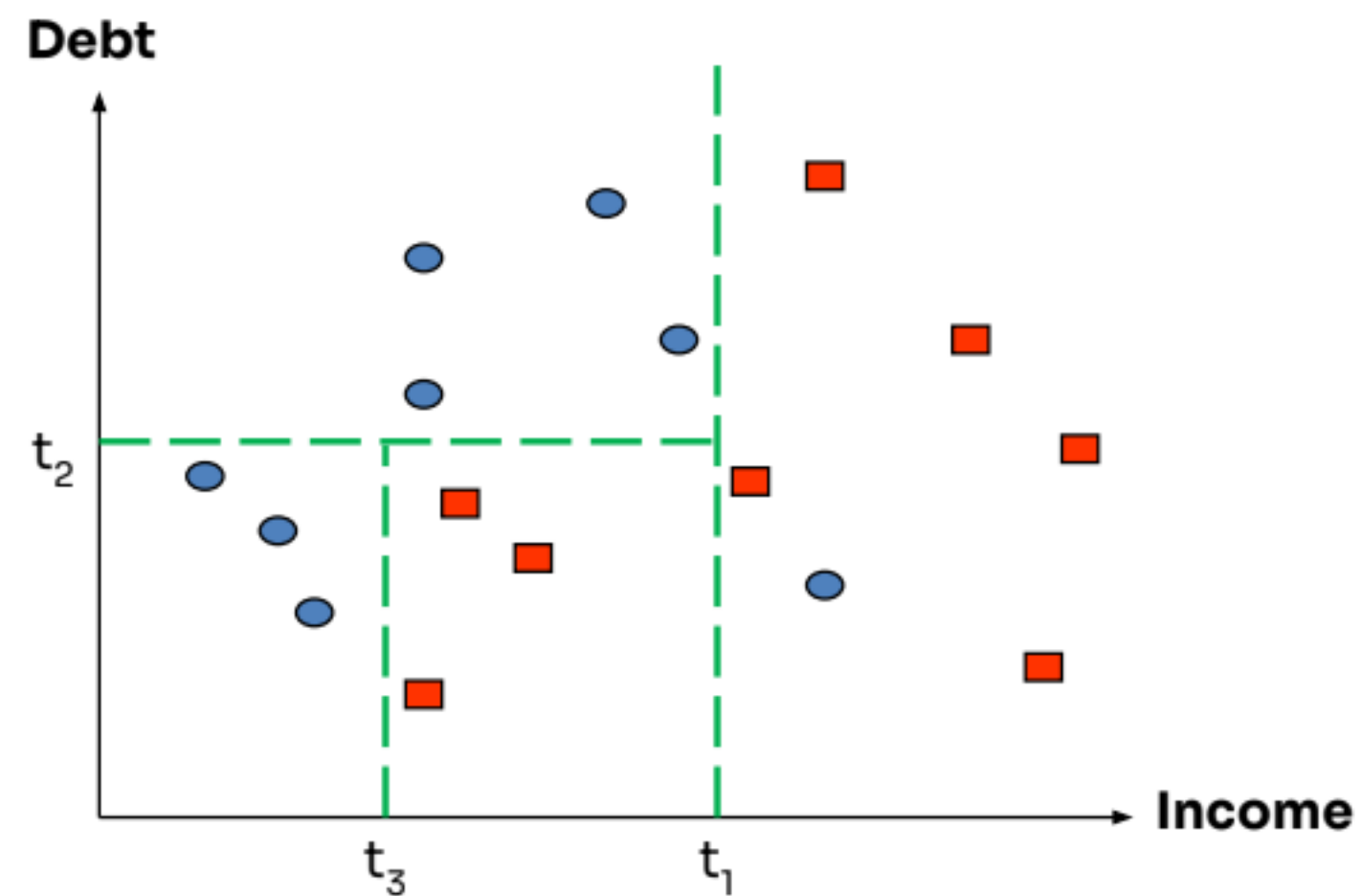
What is Decision Tree?

Decision tree membagi dataset menjadi **beberapa region** (berbentuk kotak)



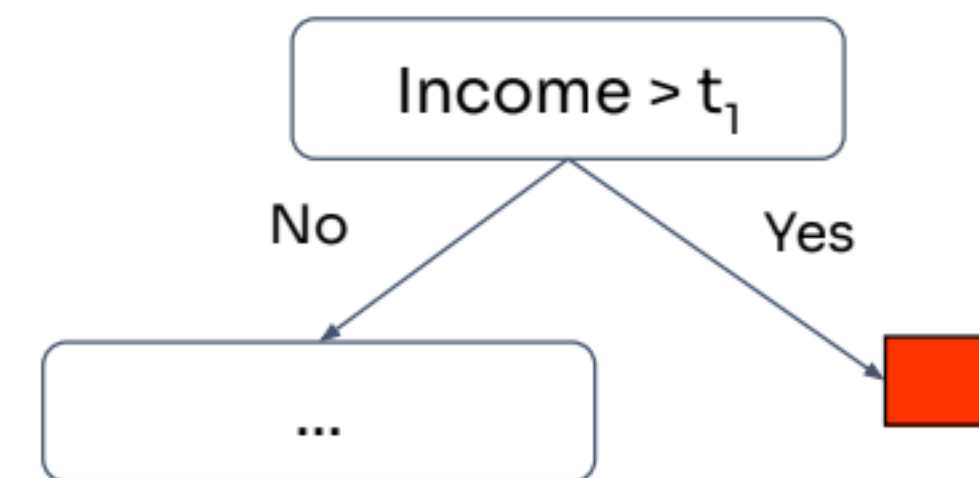
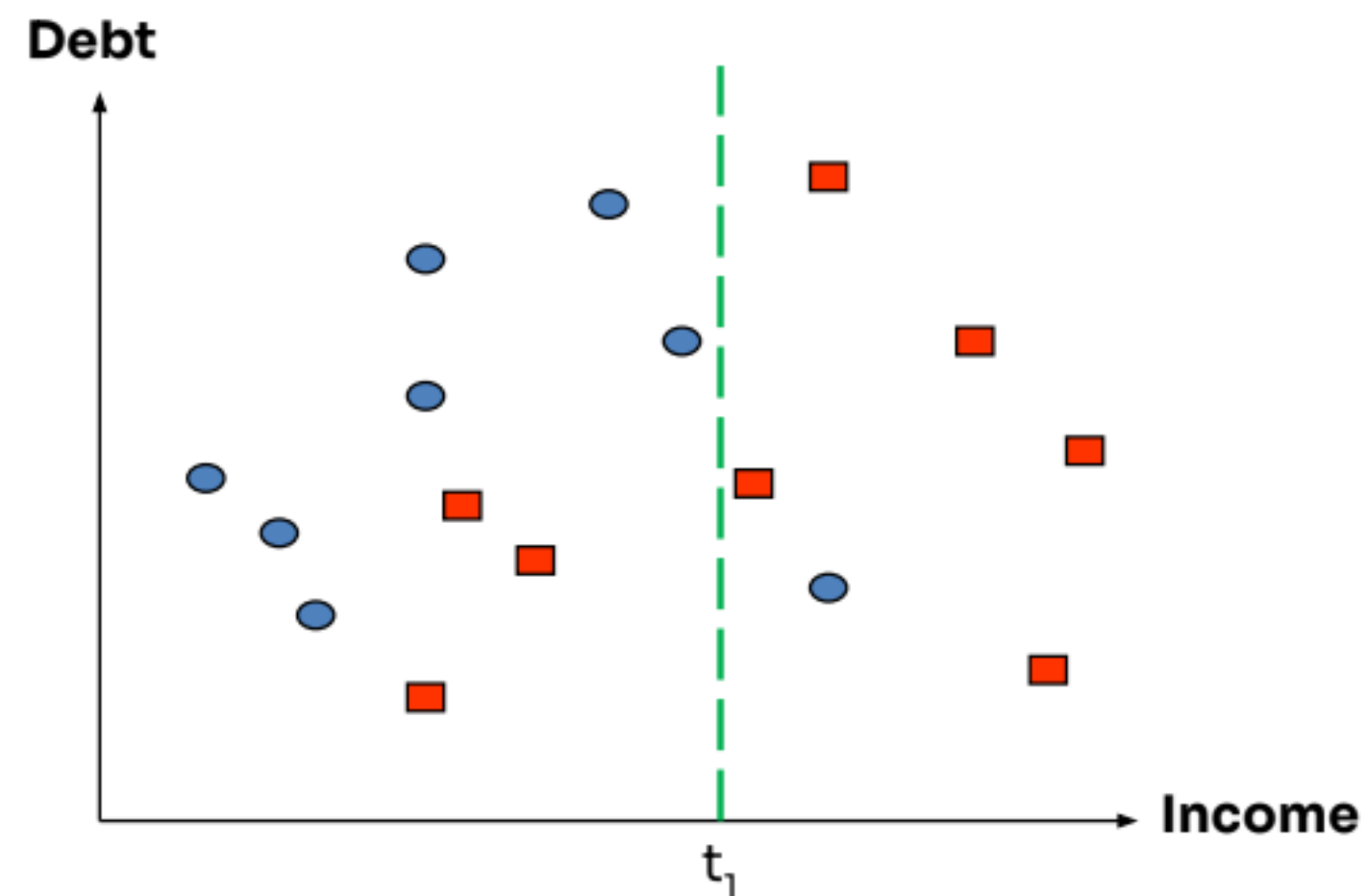
What is Decision Tree?

Disaat yang sama, decision tree dibuat.



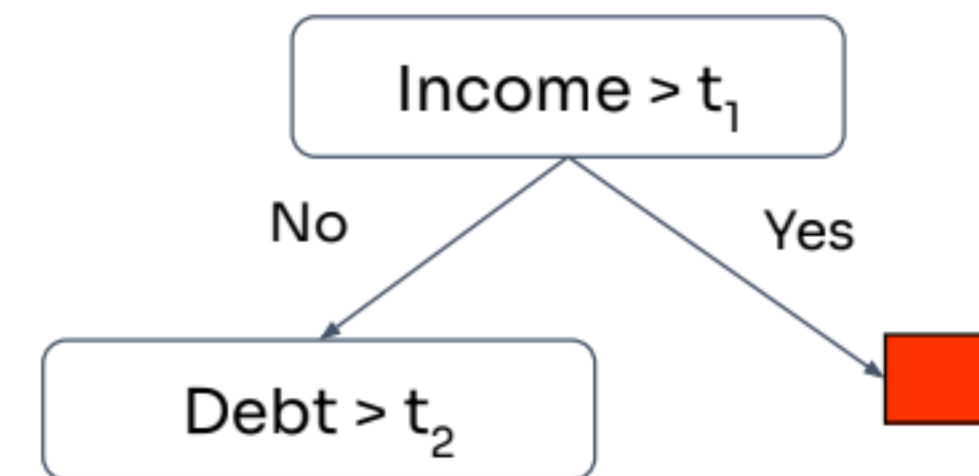
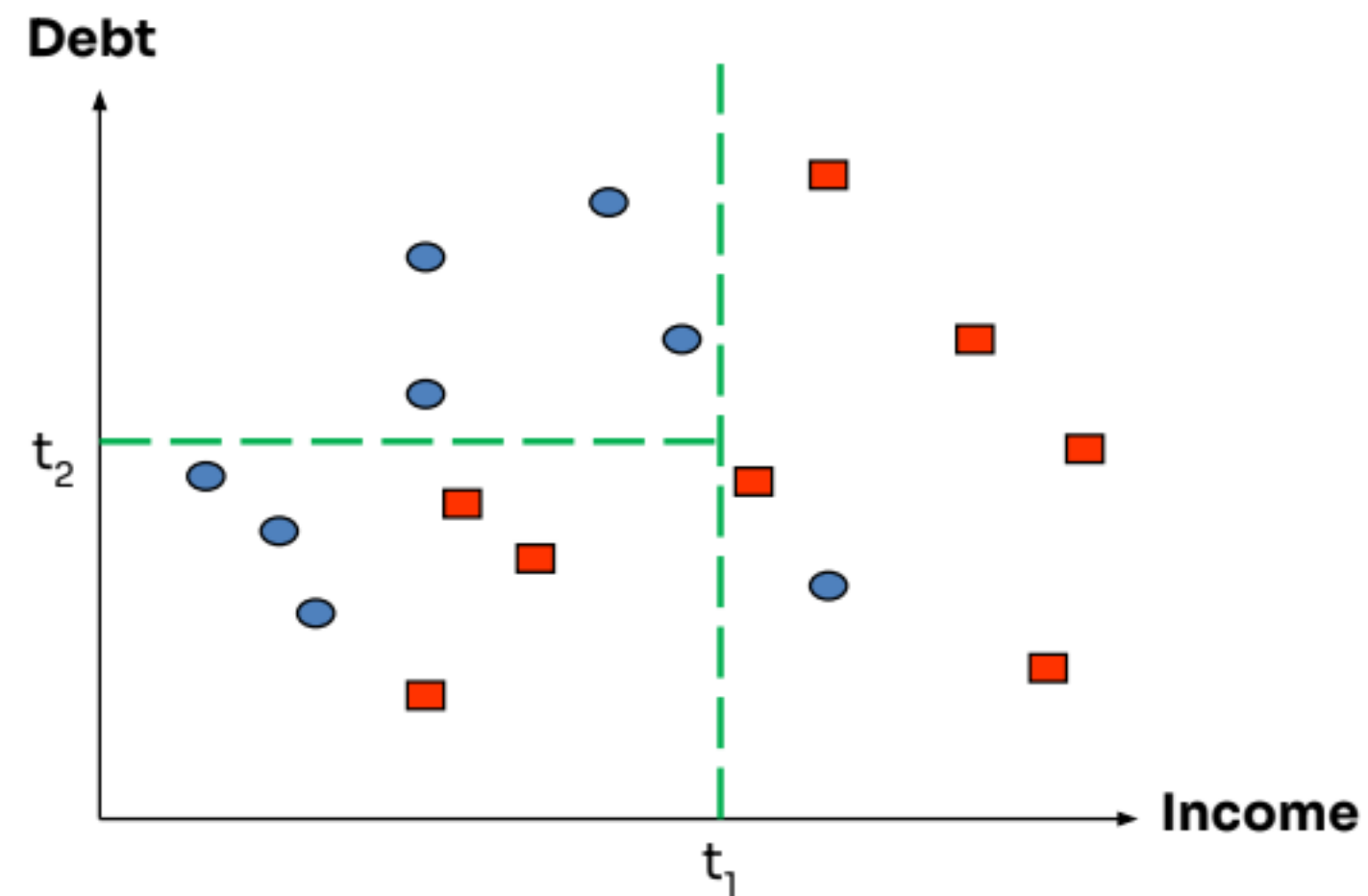
What is Decision Tree?

Pilih fitur & threshold (t_1) yang membagi target data dengan baik.



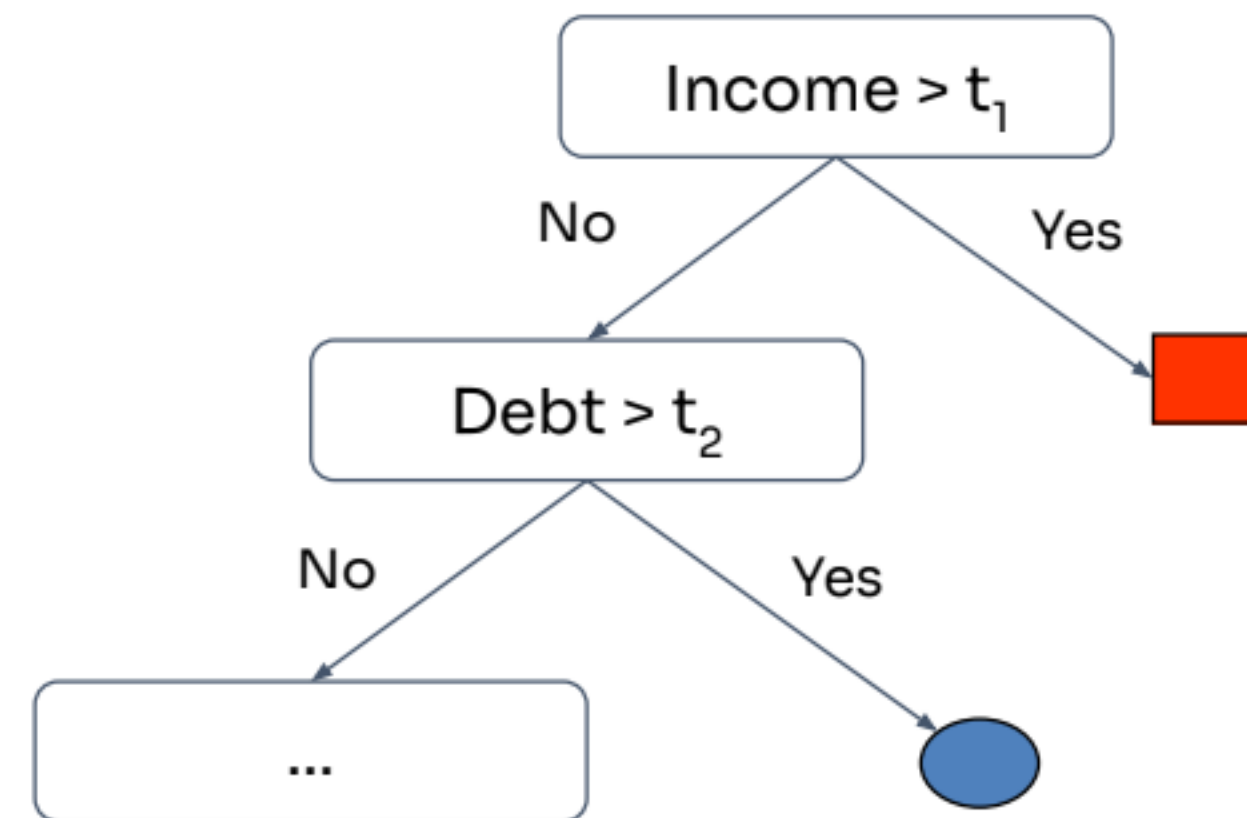
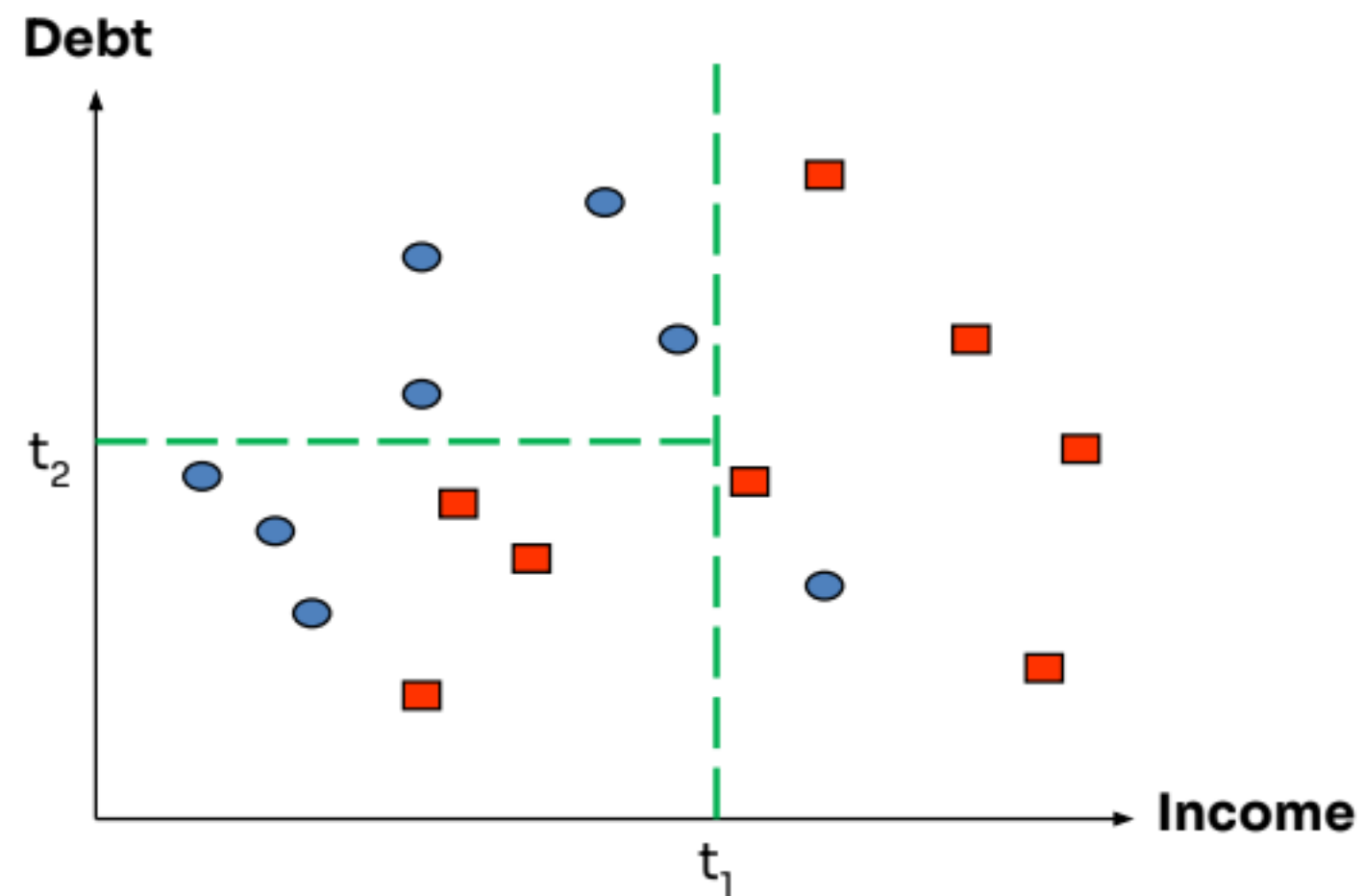
What is Decision Tree?

Ulangi kembali langkah tersebut



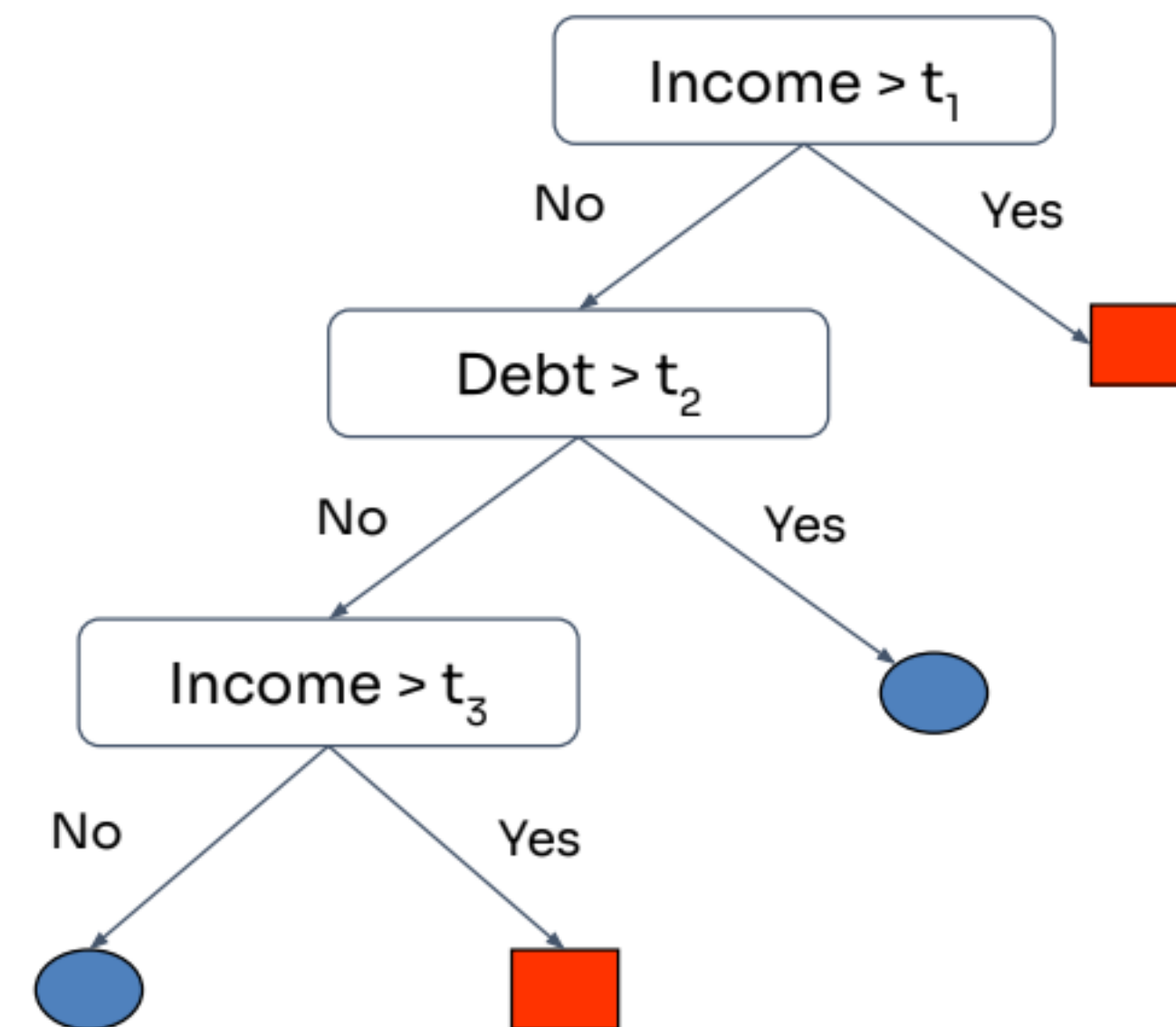
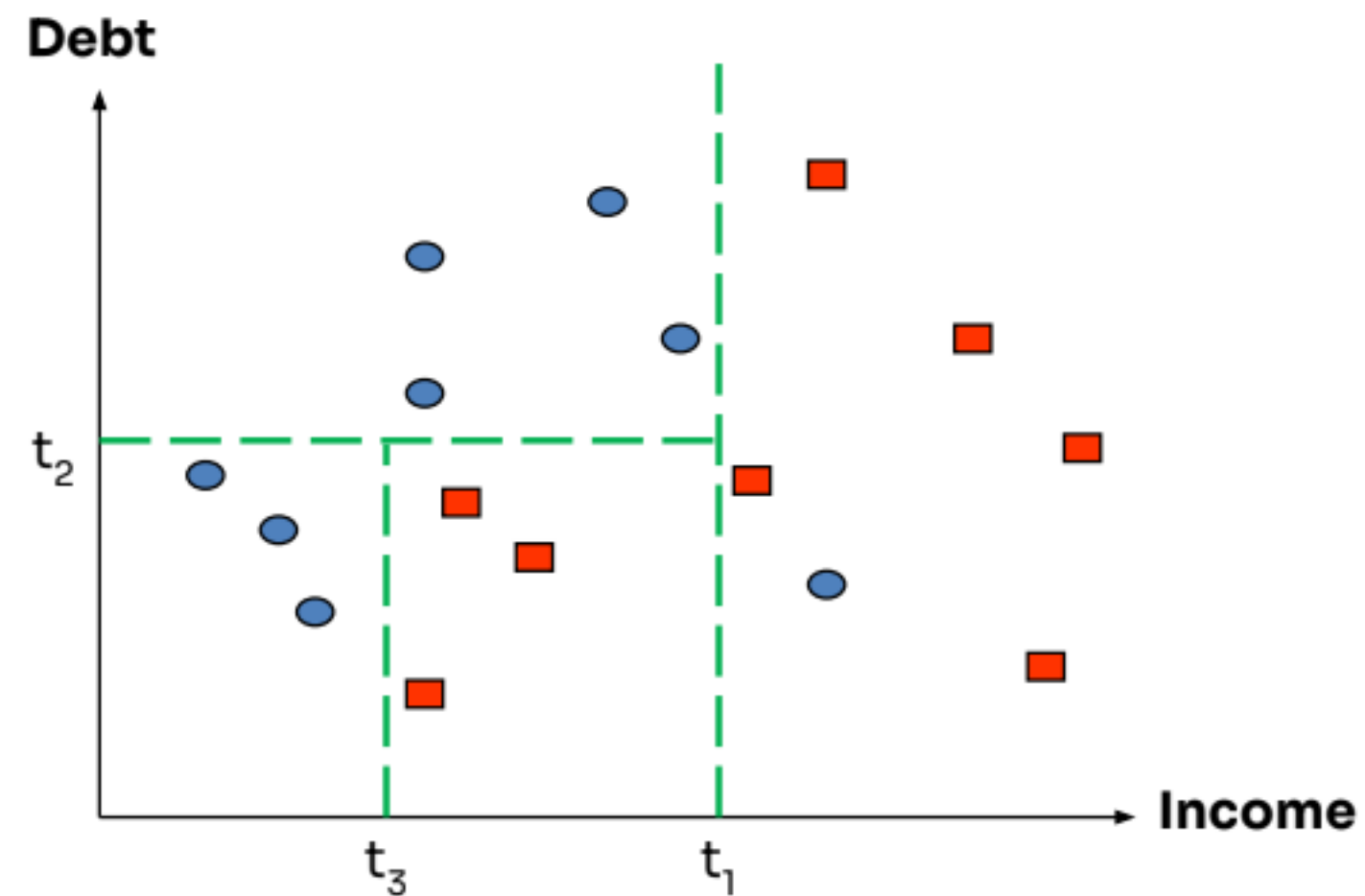
What is Decision Tree?

Ulangi kembali langkah tersebut



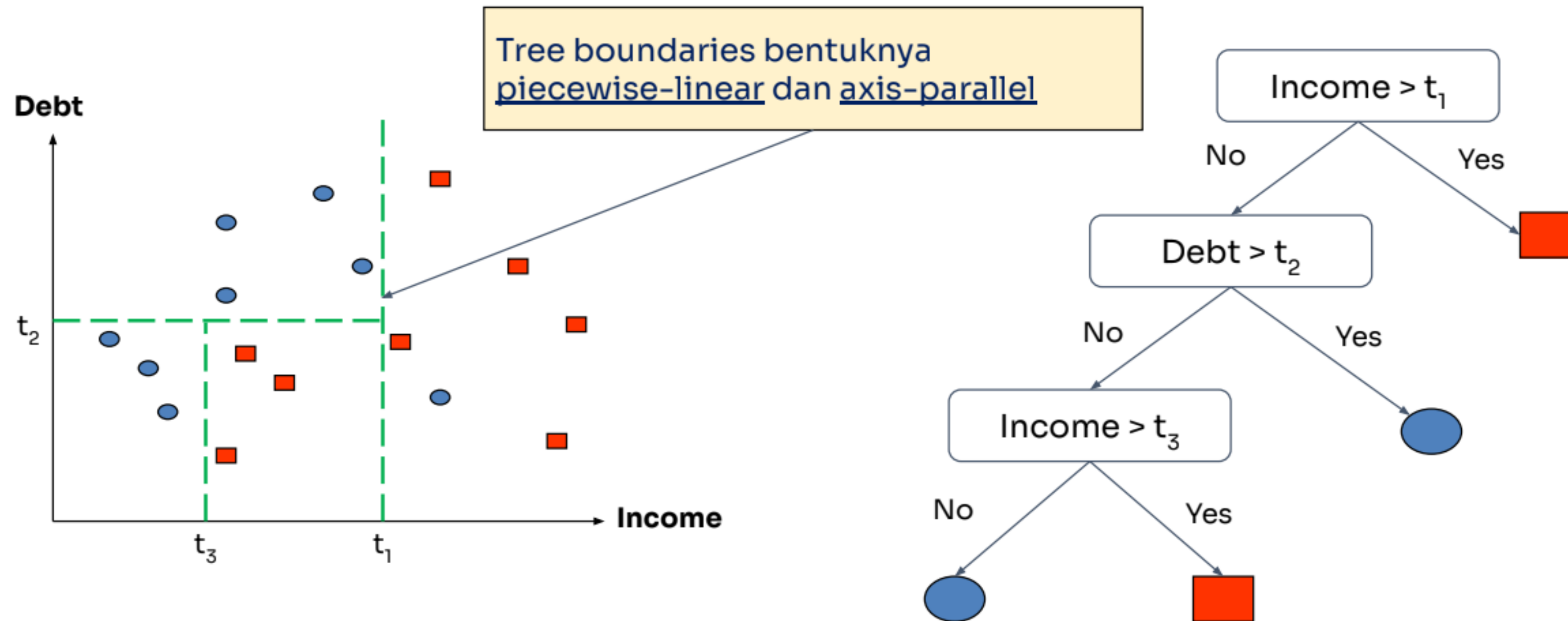
What is Decision Tree?

Ulangi kembali langkah tersebut



What is Decision Tree?

Ulangi kembali langkah tersebut

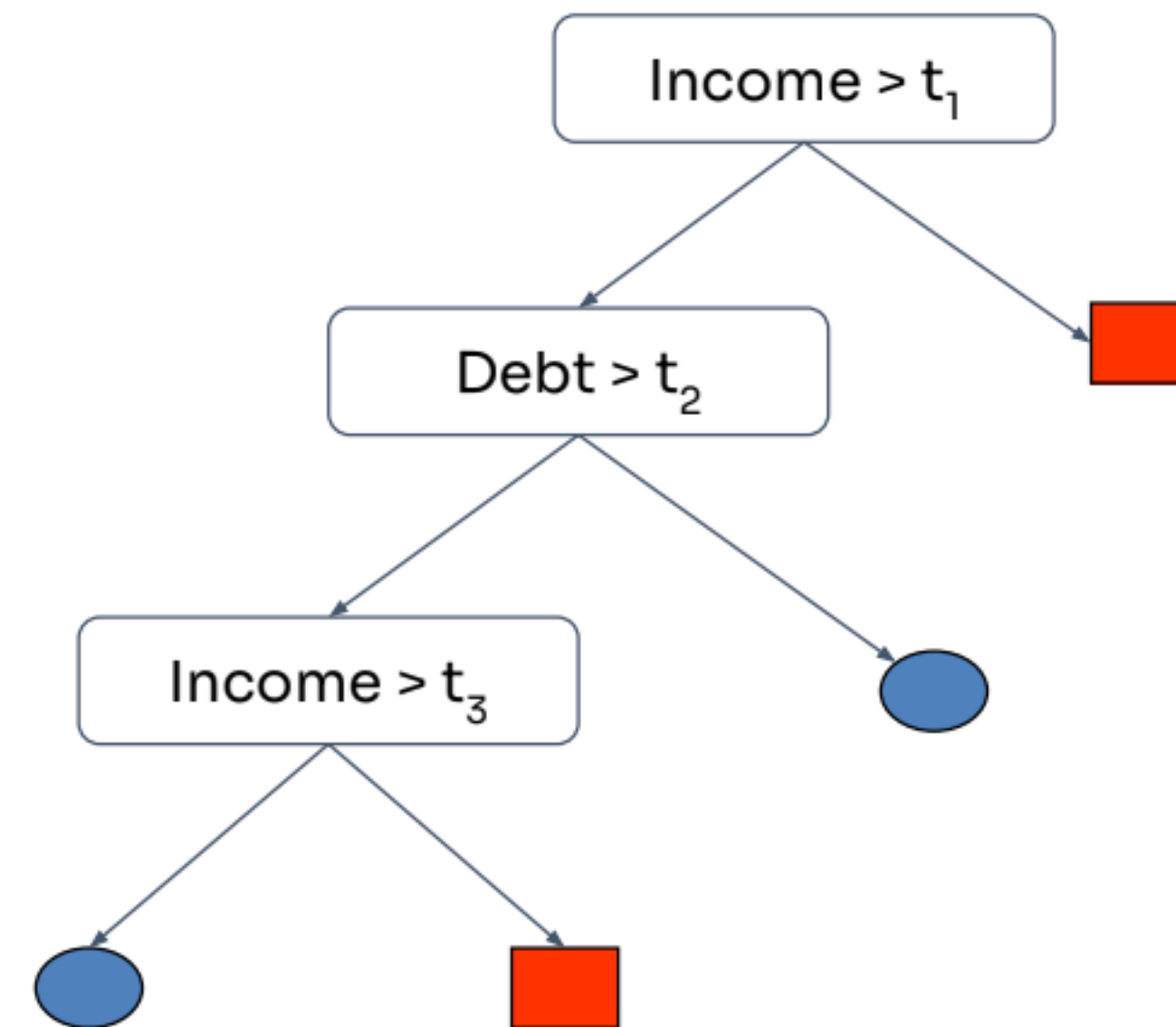


Component of Decision Tree

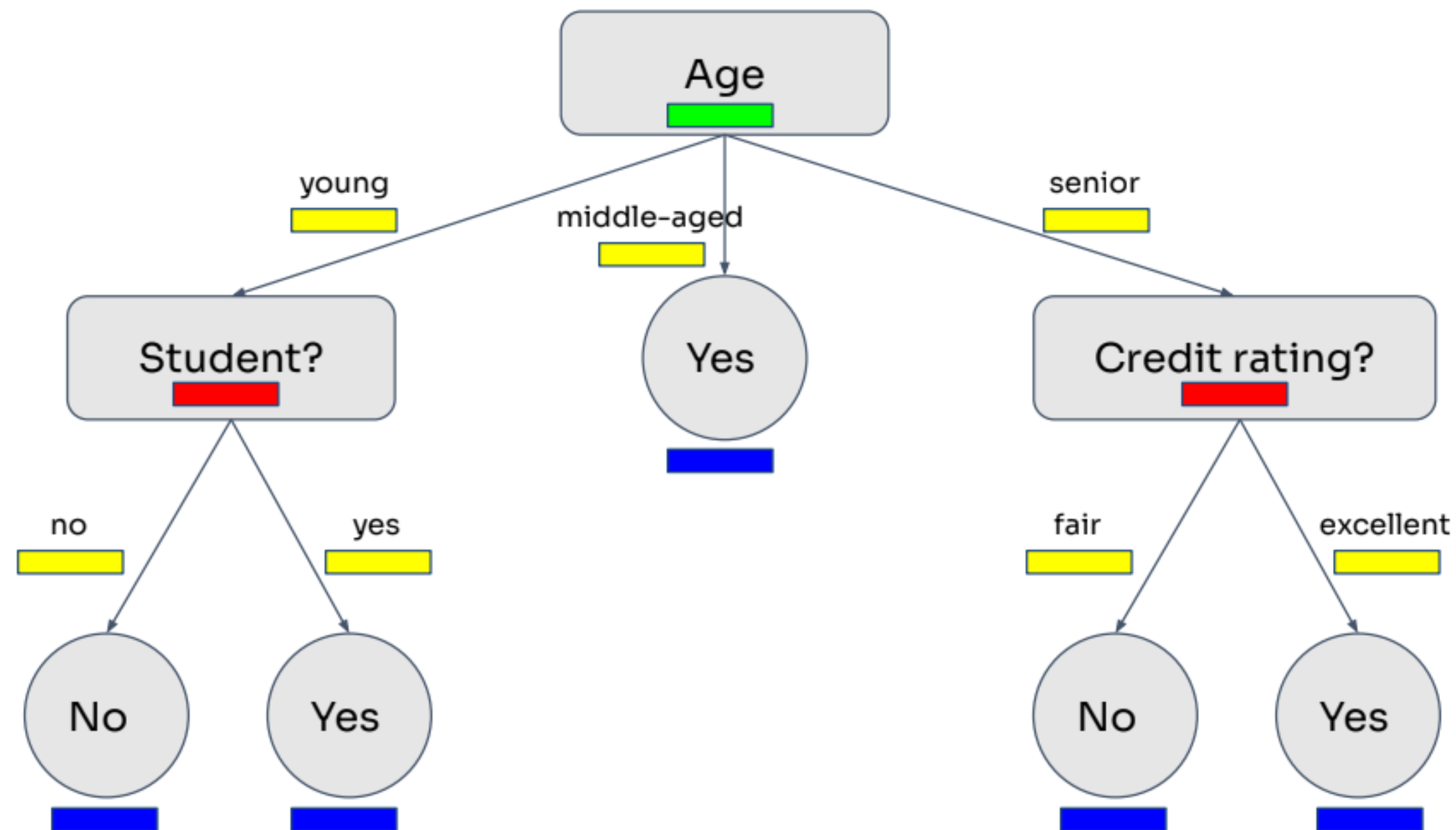
Component of Decision Tree

Komponen pada decision tree:

- Decision/internal nodes
- Leaf nodes
- Branch
- Root node



Component of Decision Tree



- Root node
- Decision/internal node
- Edge/branch
- Leaf node

Component of Decision Tree

- Decision tree seperti pohon, namun terbalik. **Root** ada di atas.
- **Decision/internal node**: uji nilai dari salah satu atribut/fitur input
- Kumpulan **edges/branches** dari node: label dengan nilai atribut/fitur yang mungkin.
- **Leaf node**: output value.

How to Split a Node

How to Split a Node: Single Variable

- Asumsikan kita punya variabel **income** dan ingin memprediksi **debt**

Feature Space dari variabel **income**

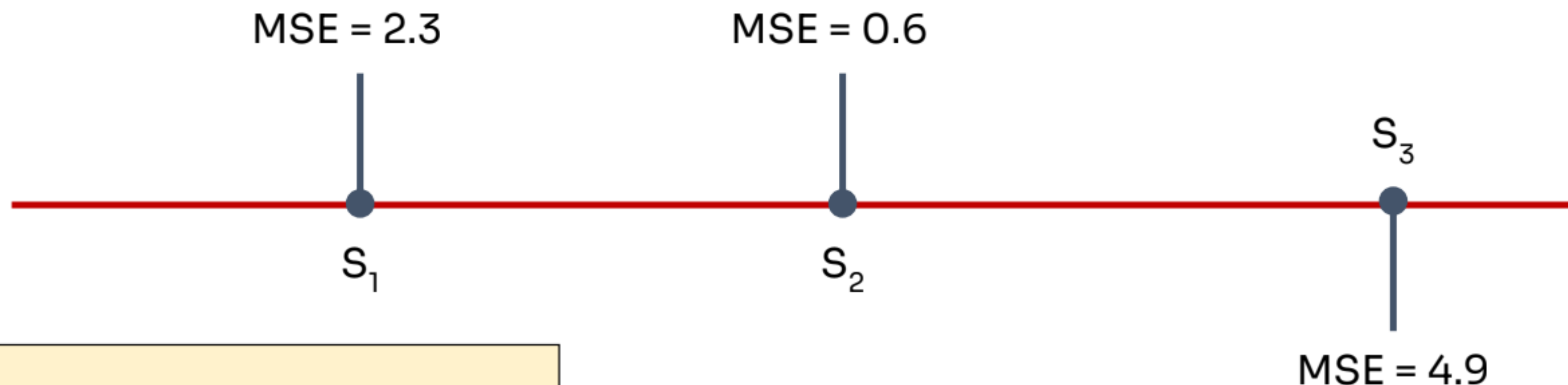
Min : 2.5 juta



Max : 15 juta

How to Split a Node: Single Variable

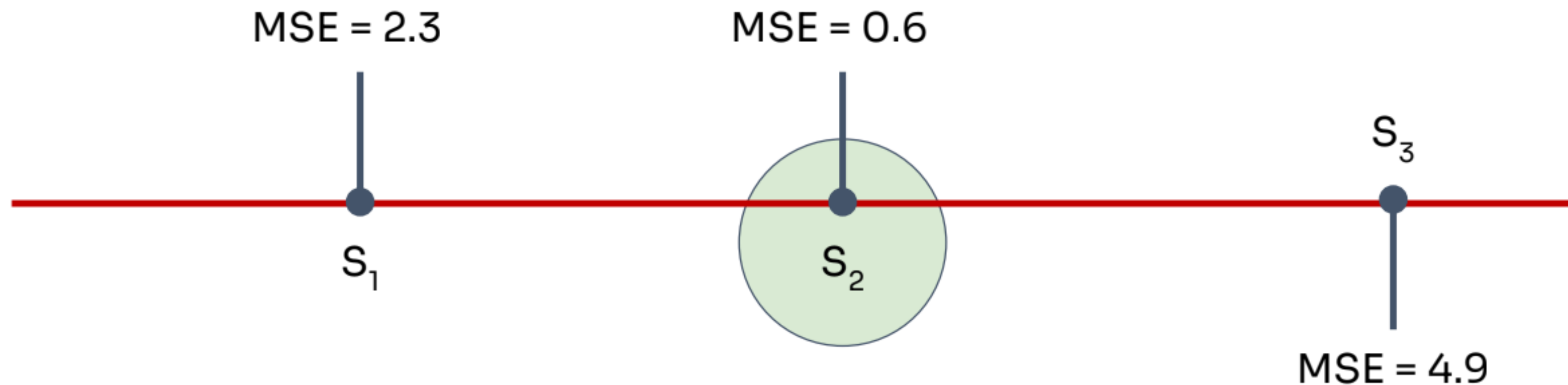
- Asumsikan kita punya variabel **income** dan ingin memprediksi **debt**
- Hitung **mean square error** di setiap angka pada feature space



Mana yang dipilih?

How to Split a Node: Single Variable

- Asumsikan kita punya variabel **income** dan ingin memprediksi **debt**
- Hitung **mean square error** di setiap angka pada feature space



Pertanyaan

- Ada berapa banyak cara untuk melakukan *splitting*?
- Bagaimana kalau kita memiliki **banyak variabel** (atribut/fitur)?

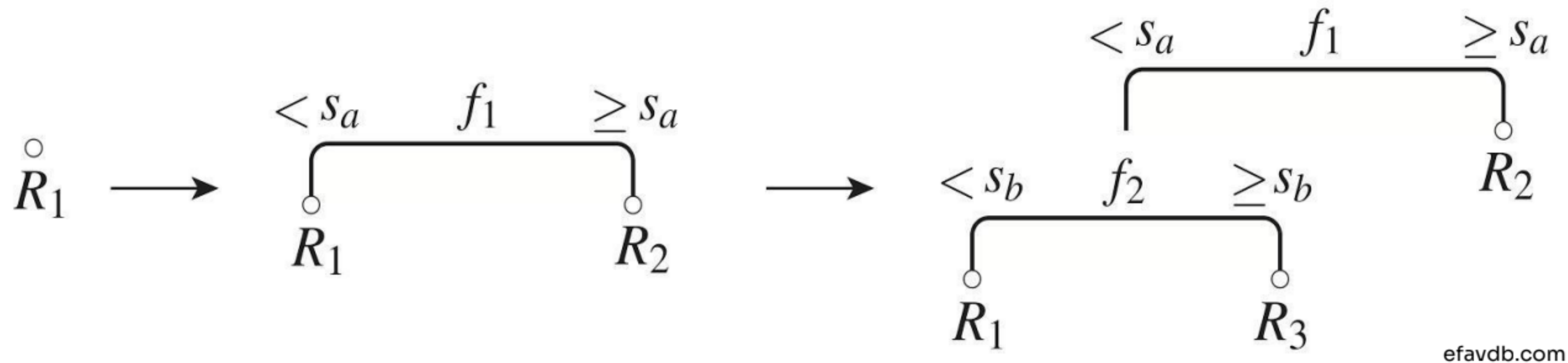
Choosing variable to split

Variable 1	Split in S_1	MSE = 3,143
Variable 2	Split in S_2	MSE = 1,433
Variable 3	Split in S_3	MSE = 4,331
$\vdots$	$\vdots$	$\vdots$
Variable k	Split in S_k	MSE = 3,314

Yang mana yang kita pilih?

Greedy Algorithm

- Tree dibentuk secara rekursif, **satu branch** tiap waktu.



- Split ditentukan pada kondisi yang paling besar menurunkan *cost function*, J
- Algoritma dihentikan ketika *stopping criterion* tercapai.

Regression & Classification

Regression	Classification
Output: • $f(\mathbf{x}): \mathbb{R}^p \rightarrow \mathbb{R}$	Output: • Binary Classification: $\Omega = \{-1, 1\}$ • Multi-class Classification: $\Omega = \{1, 2, 3, \dots, C\}$ • $f(\mathbf{x}): \mathbb{R}^p \rightarrow \Omega$

Classification Trees

Sample Dataset

- Kolom variabel → $\mathbf{X}_i$
- Observasi (baris) → $(x^{(i)}, y^{(i)})$
- Label **Class** → bermain tenis/tidak

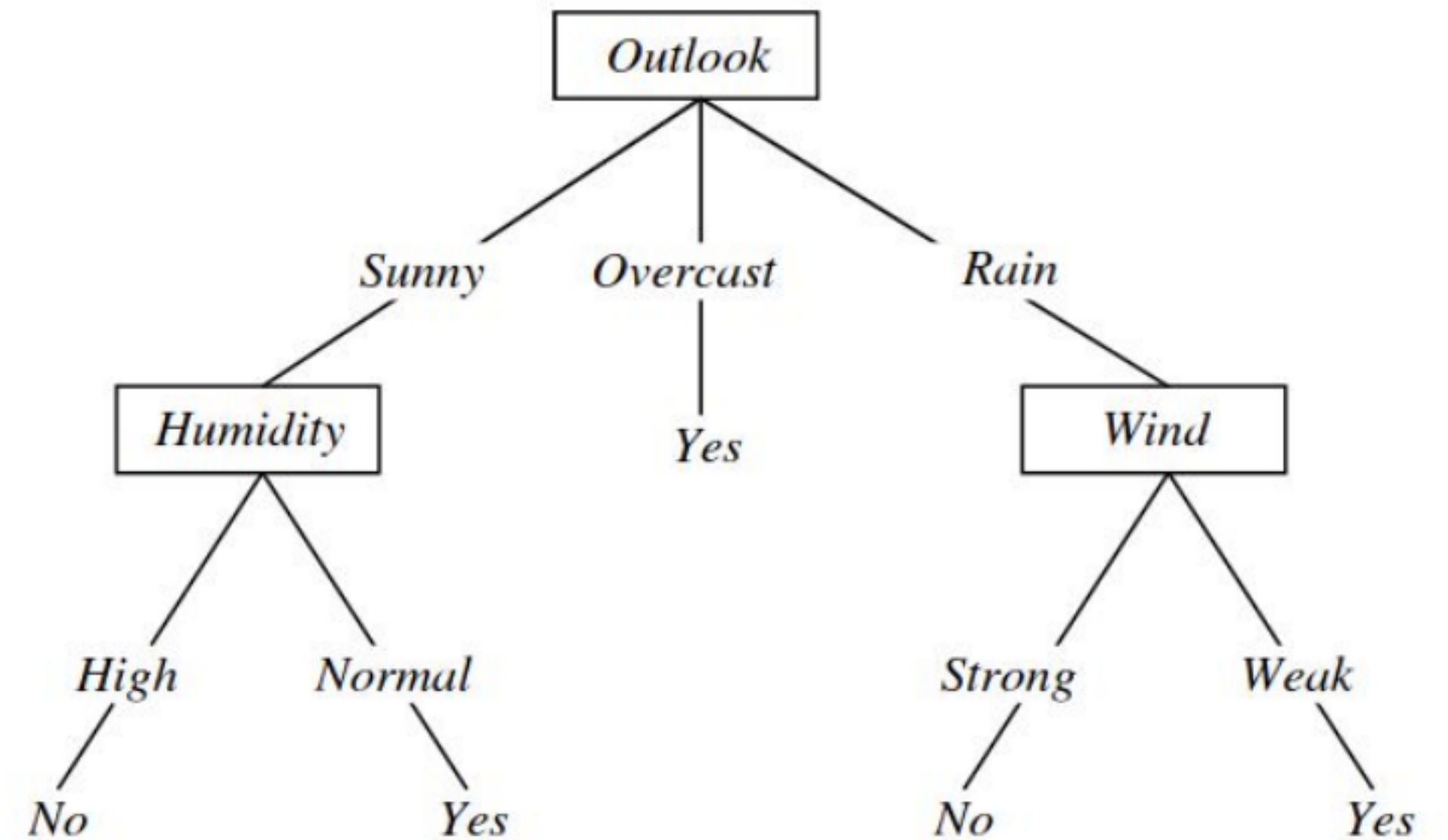
$\langle x^{(i)}, y^{(i)} \rangle$

Predictors				Response
Outlook	Temperature	Humidity	Wind	Class
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

based on slide by Tom Mitchell

Decision Tree

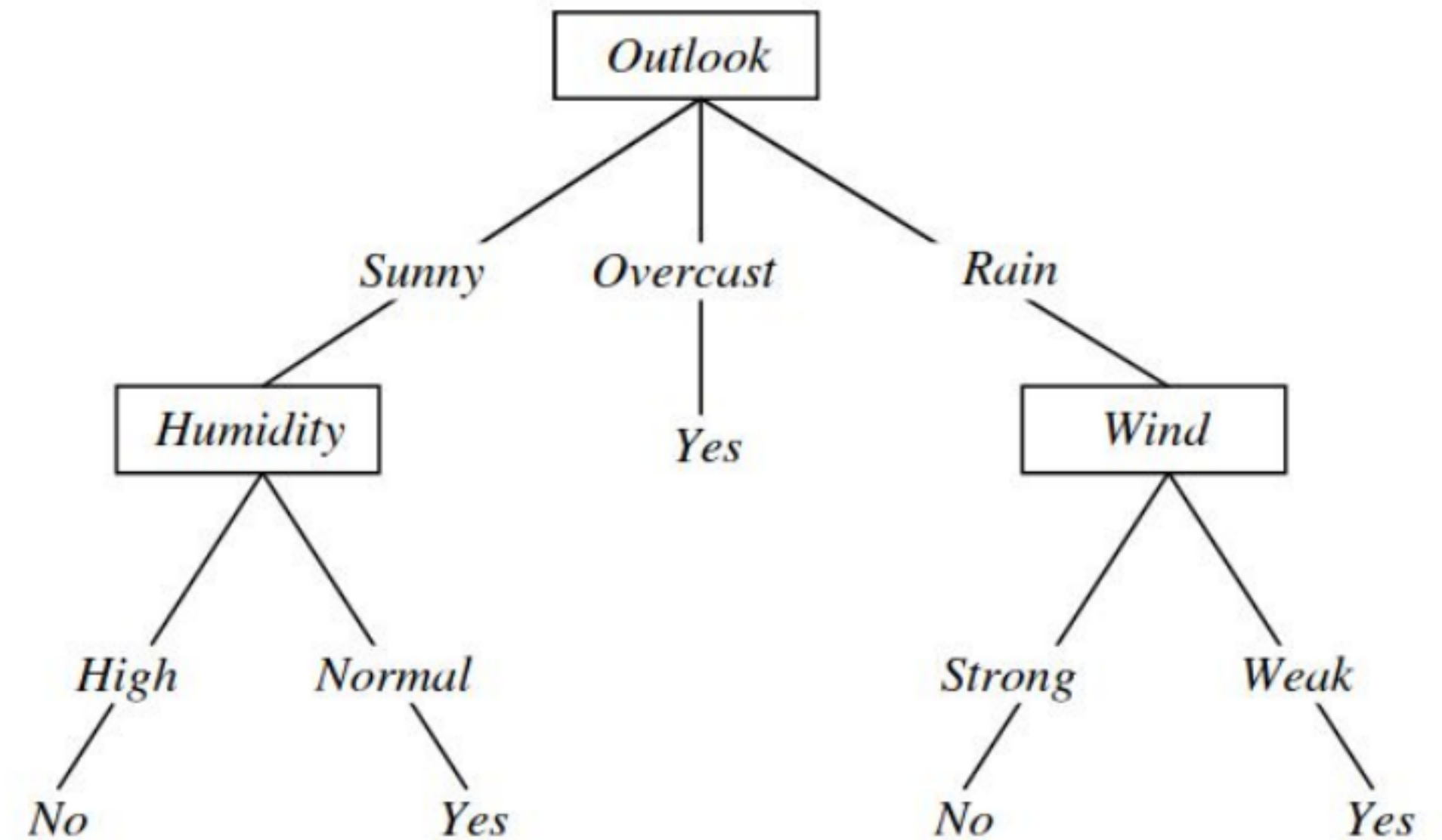
- Salah satu decision tree yang mungkin



based on slide by Tom Mitchell

Decision Tree

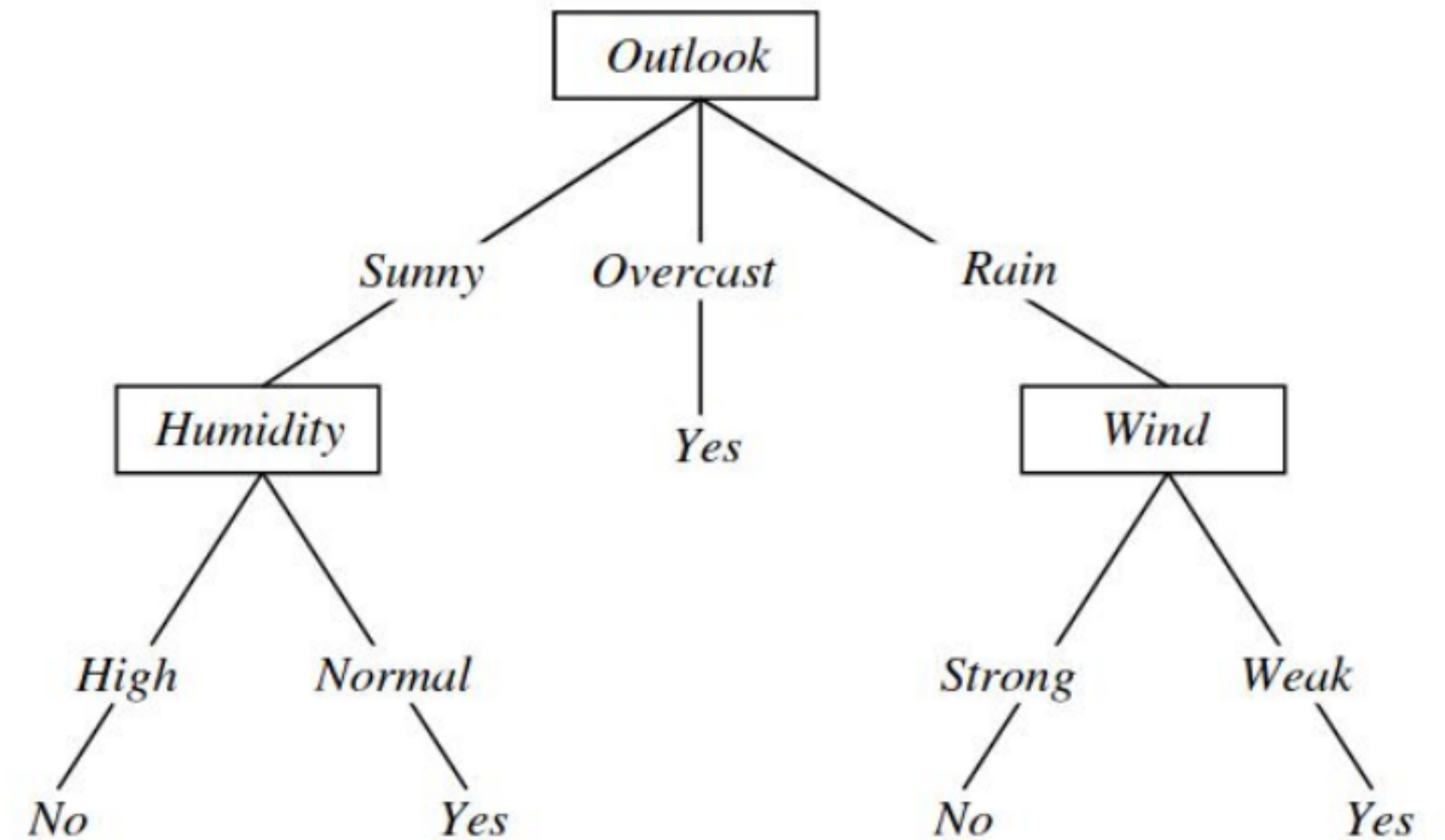
- Salah satu decision tree yang mungkin
- Uji 1 atribut di tiap internal node.
- Setiap branch, pilih 1 nilai dari X_i
- Setiap leaf, prediksi Y atau, $p(Y \mid x \in \text{leaf})$



based on slide by Tom Mitchell

Decision Tree

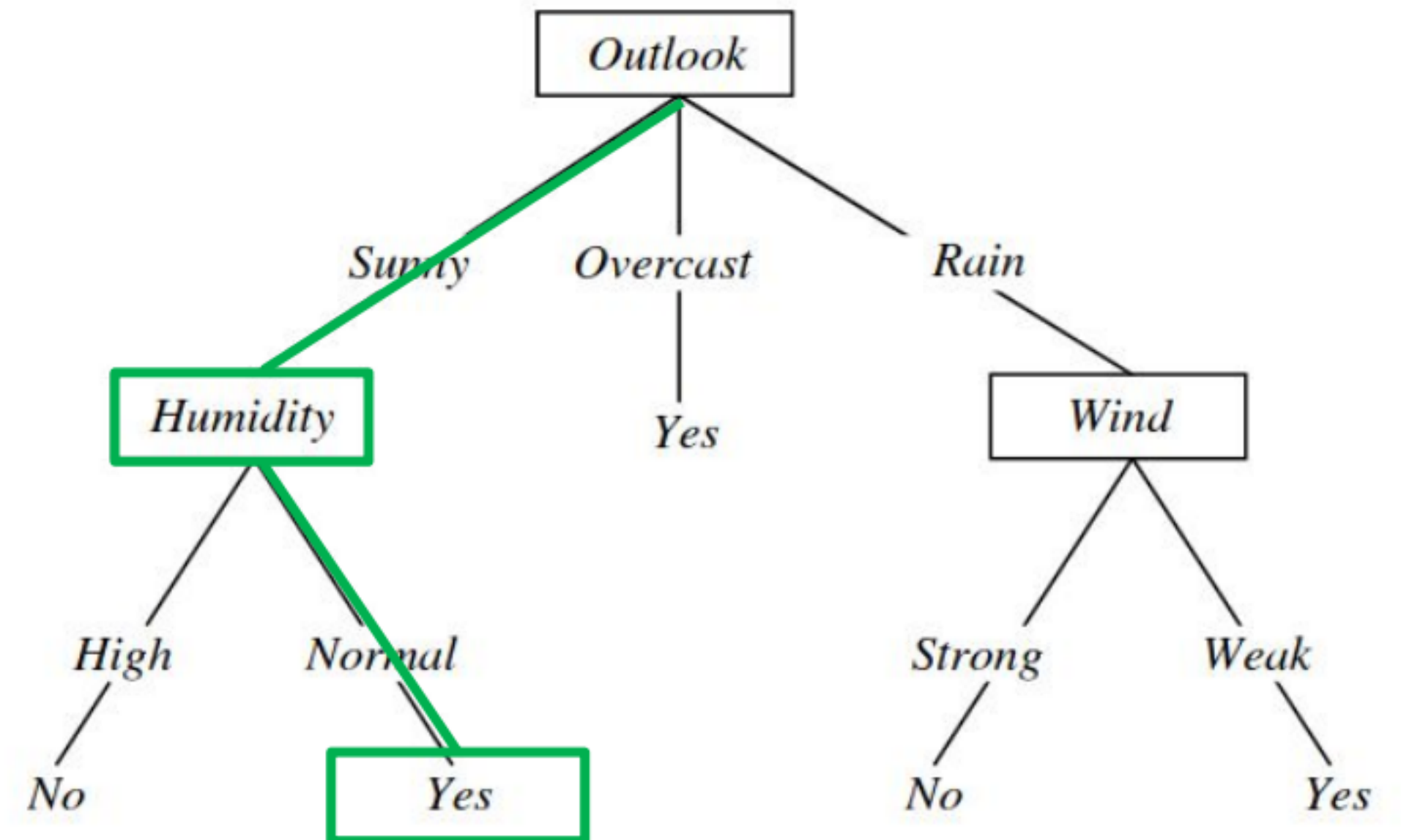
- Salah satu decision tree yang mungkin
- **Bagaimana hasil prediksi** apabila:
 - outlook : sunny
 - humidity : normal



based on slide by Tom Mitchell

Decision Tree

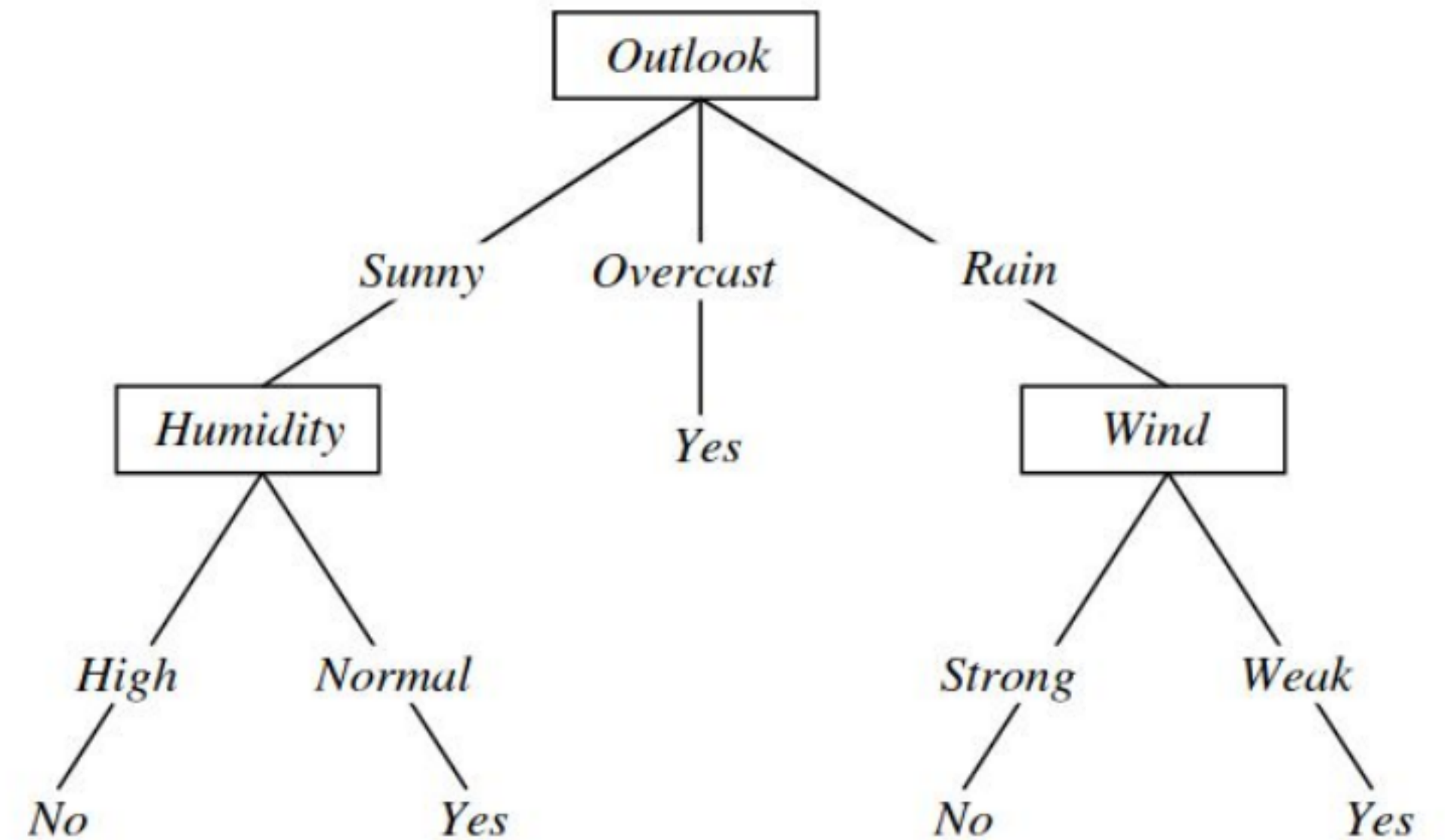
- Salah satu decision tree yang mungkin
- **Bagaimana hasil prediksi** apabila:
 - outlook : sunny
 - humidity : normal



based on slide by Tom Mitchell

Decision Tree

- Salah satu decision tree yang mungkin
- **Bagaimana hasil prediksi** apabila:
 - outlook : sunny
 - temperature : hot
 - humidity : high
 - wind : weak

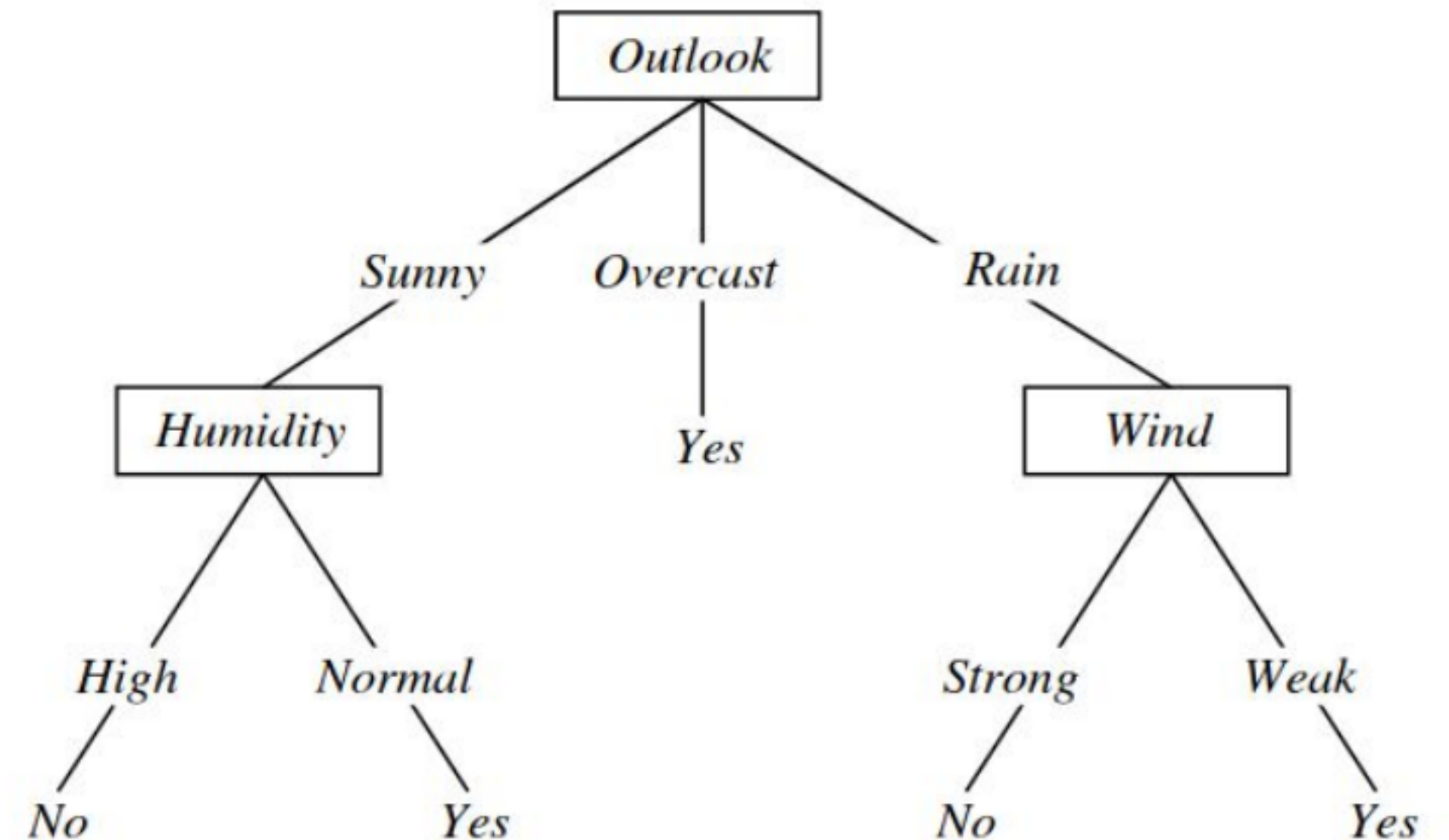


based on slide by Tom Mitchell

Decision Tree

- Salah satu decision tree yang mungkin
- **Bagaimana hasil prediksi** apabila:
 - outlook : sunny
 - temperature : hot
 - humidity : high
 - wind : weak
- Hasil prediksi “yes” ketika

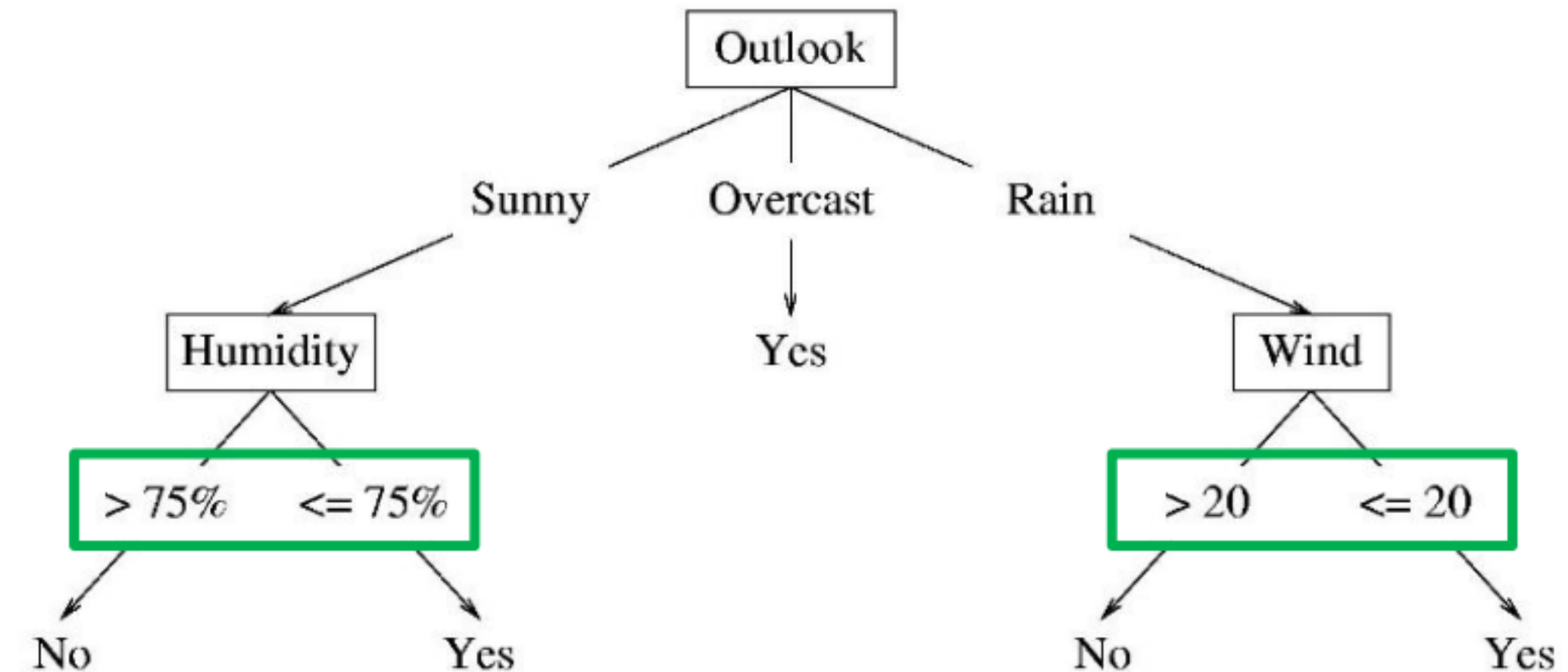
(outlook = “sunny” **dan** humidity = “normal”)
atau
 (outlook = “overcast”)
atau
 (outlook = “rain” **dan** wind = “weak”)



based on slide by Tom Mitchell

Decision Tree

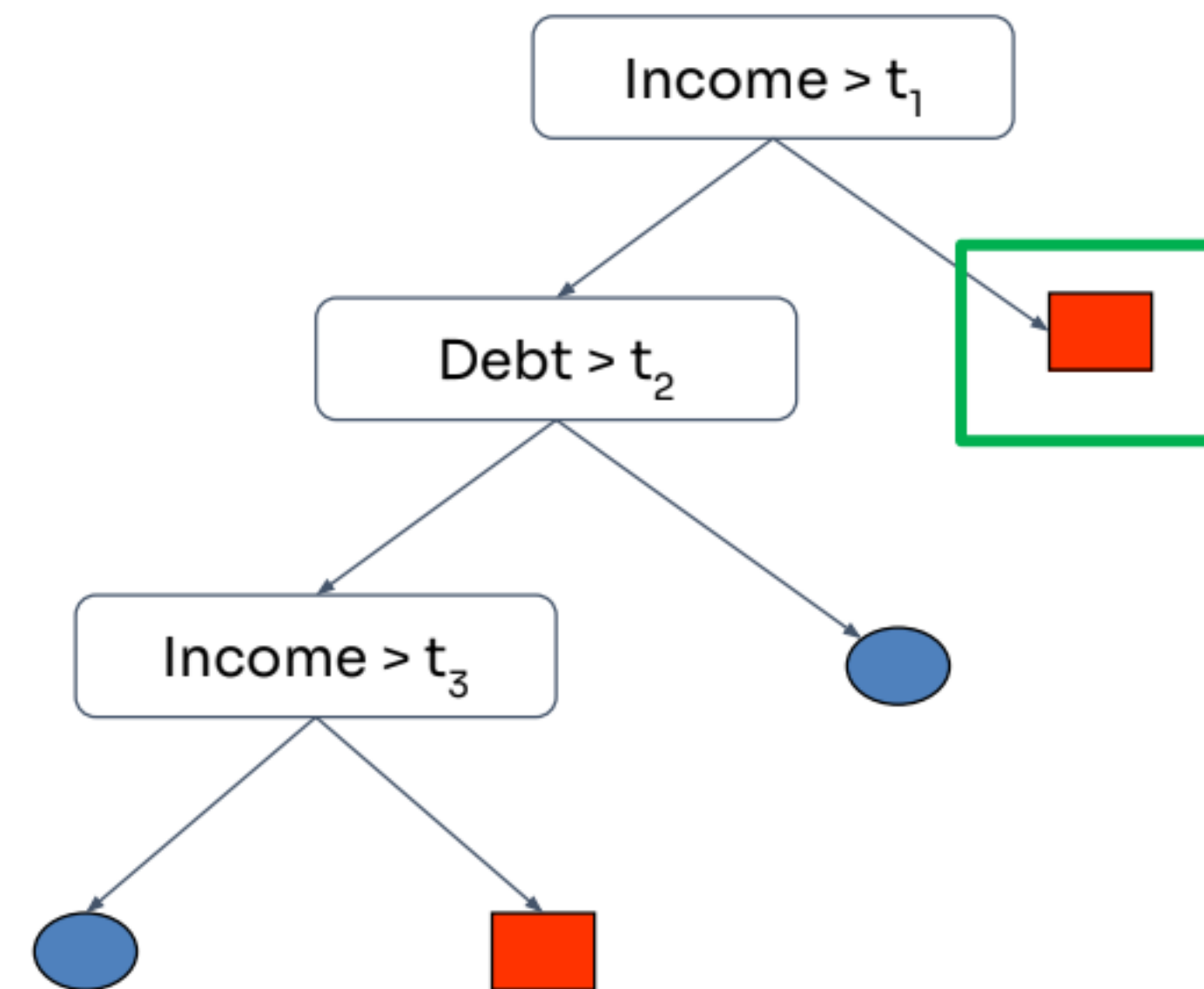
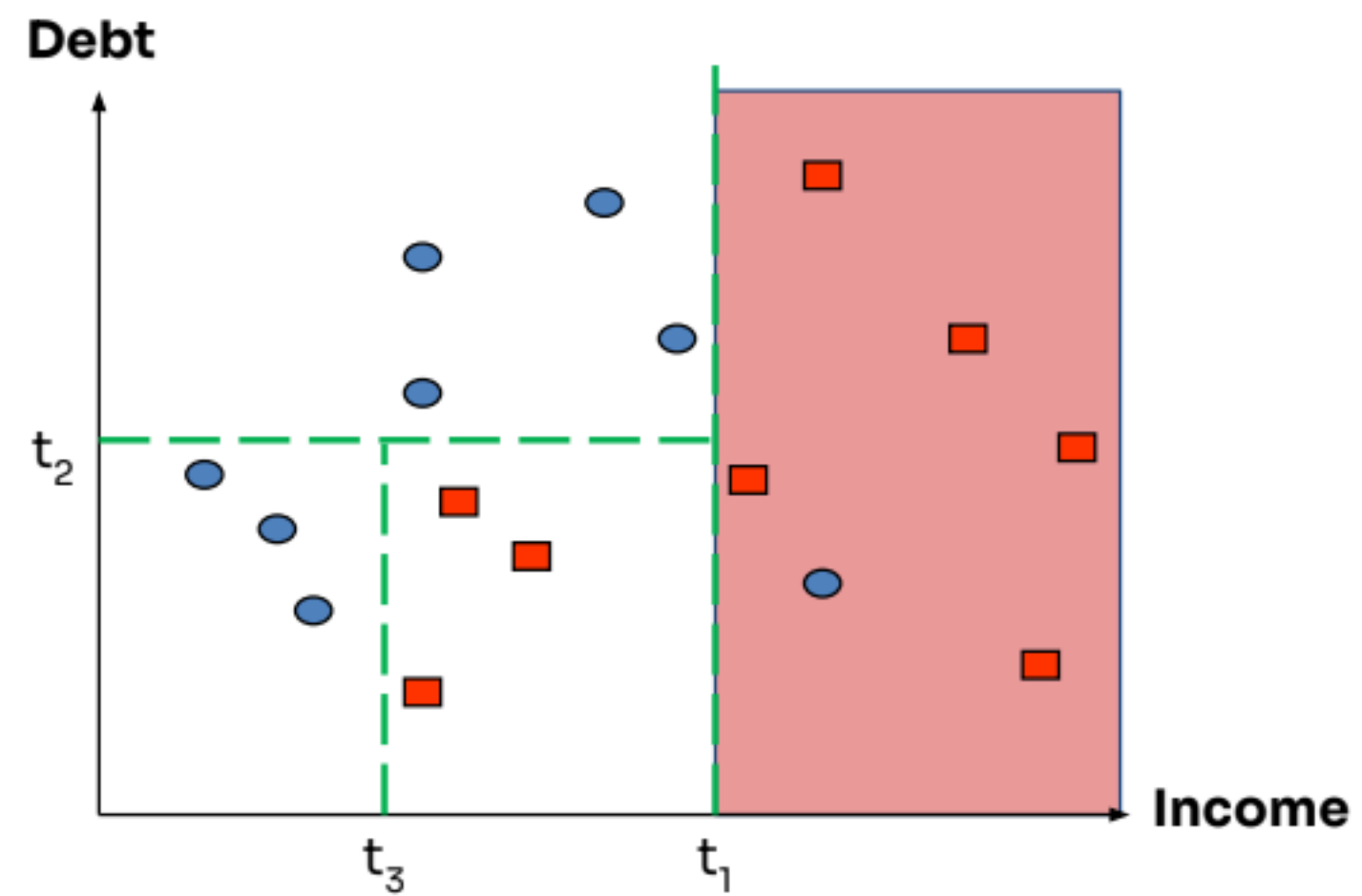
- Jika fitur kontinu, **internal nodes** dapat melakukan uji nilai fitur dengan **threshold**



based on slide by Tom Mitchell

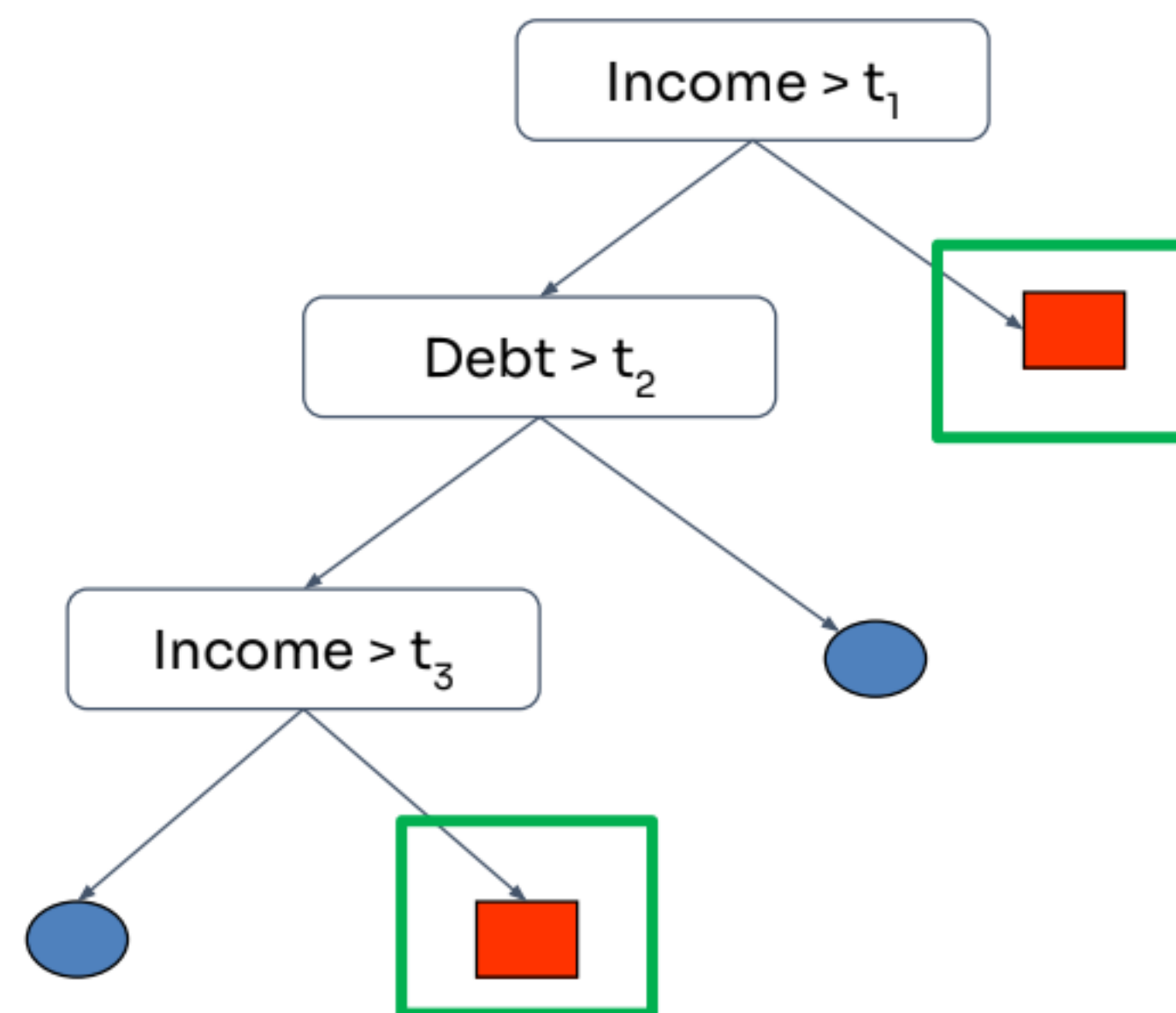
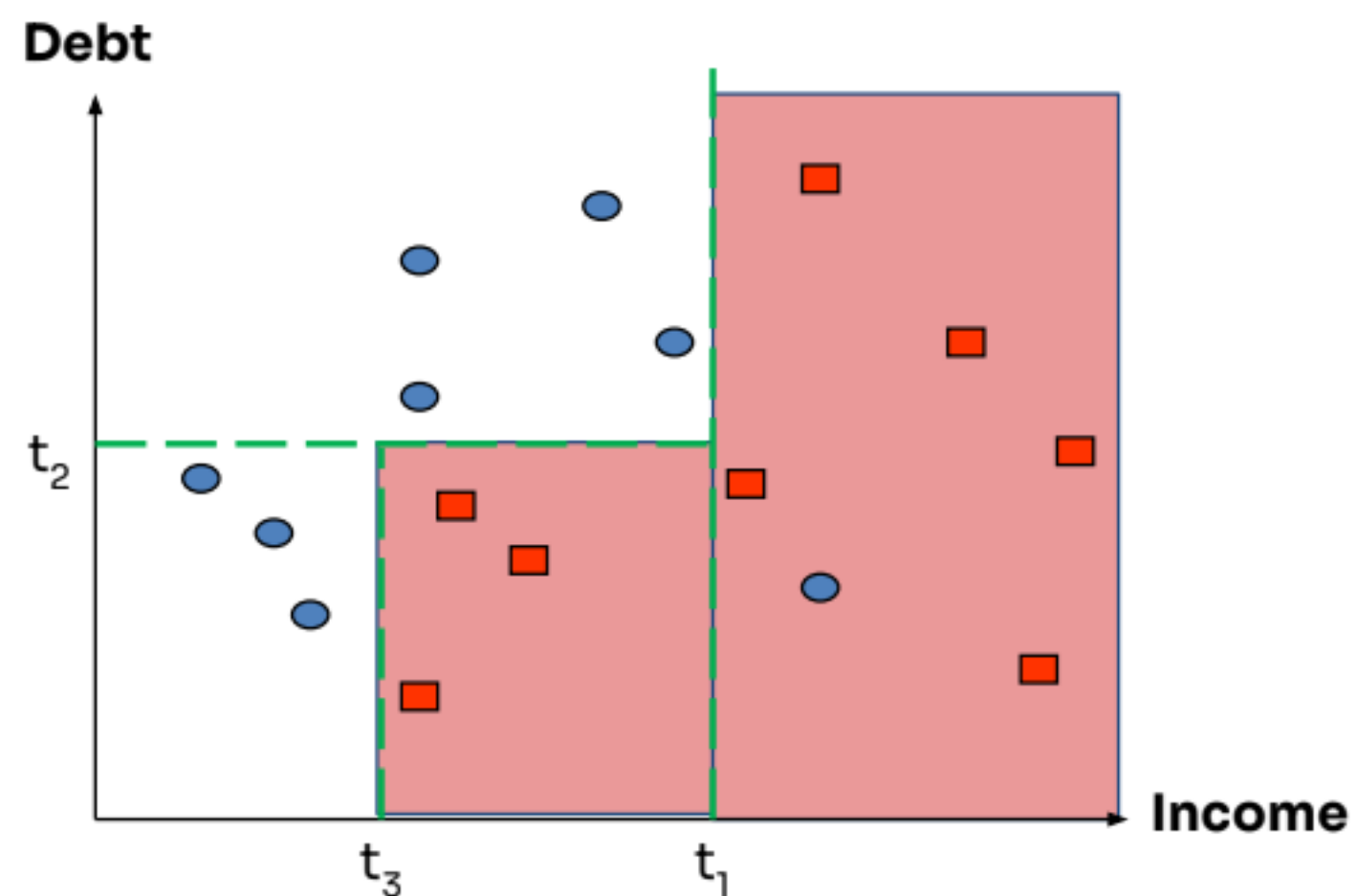
Decision Tree

Decision boundary



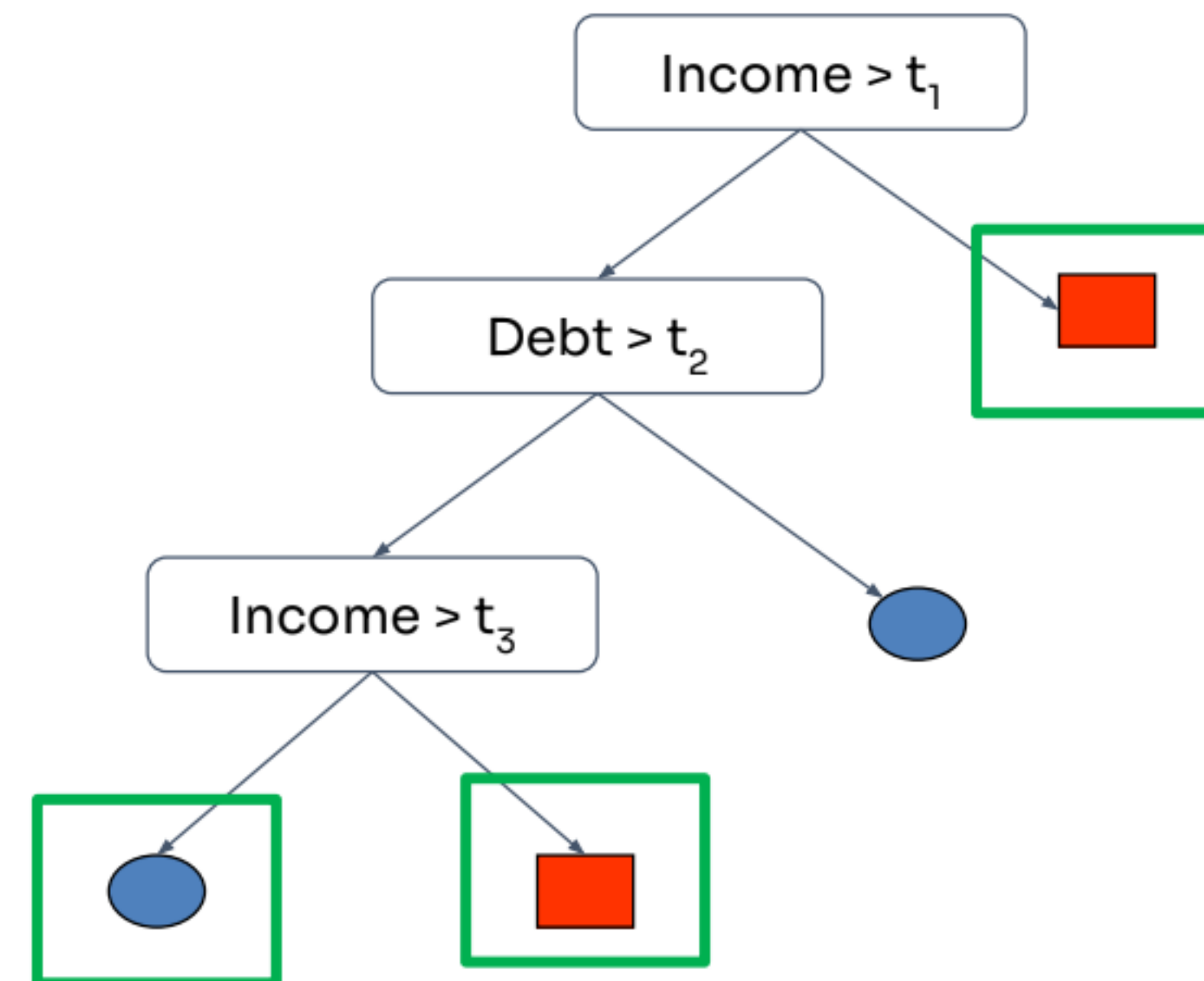
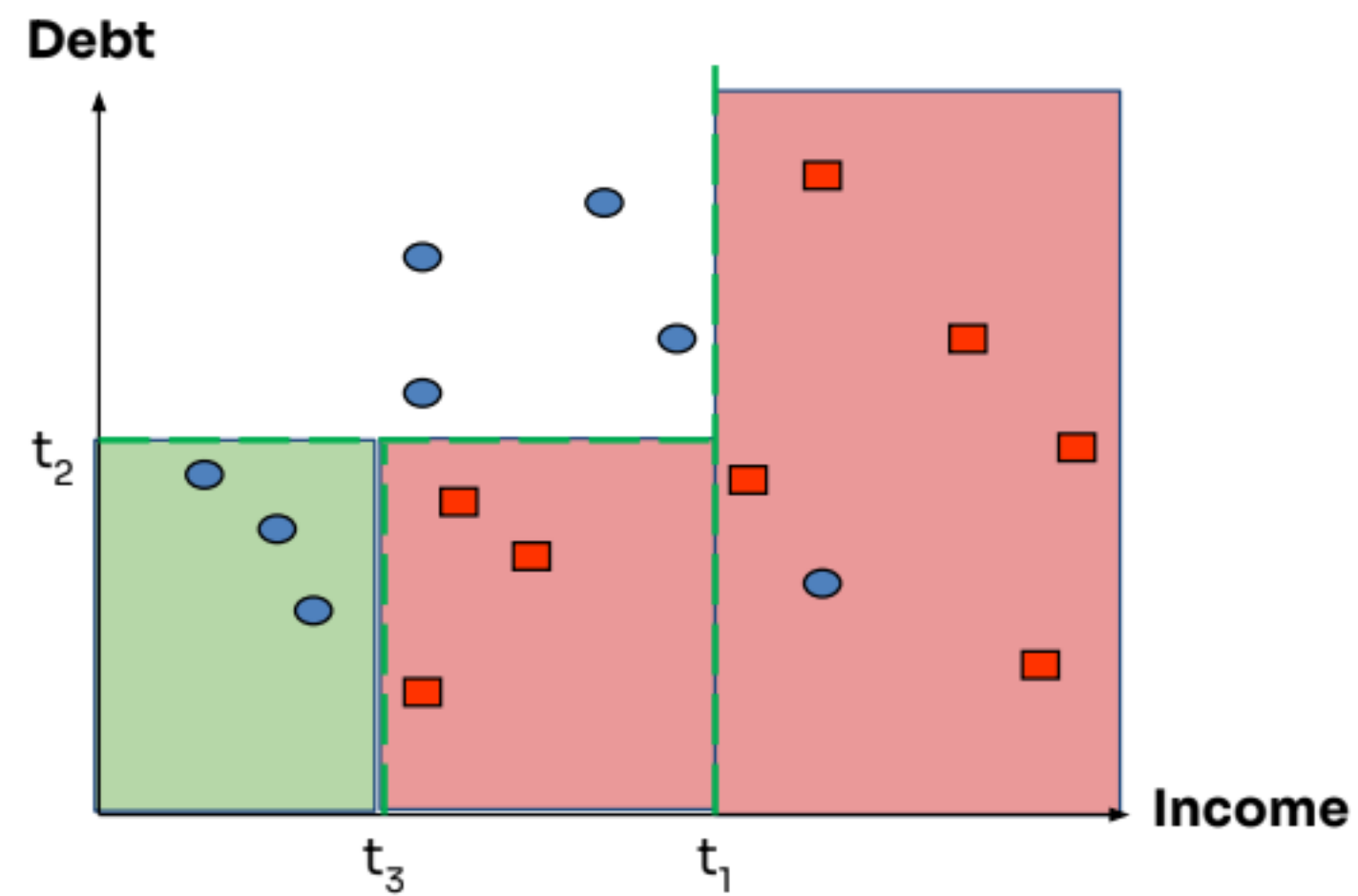
Decision Tree

Decision boundary



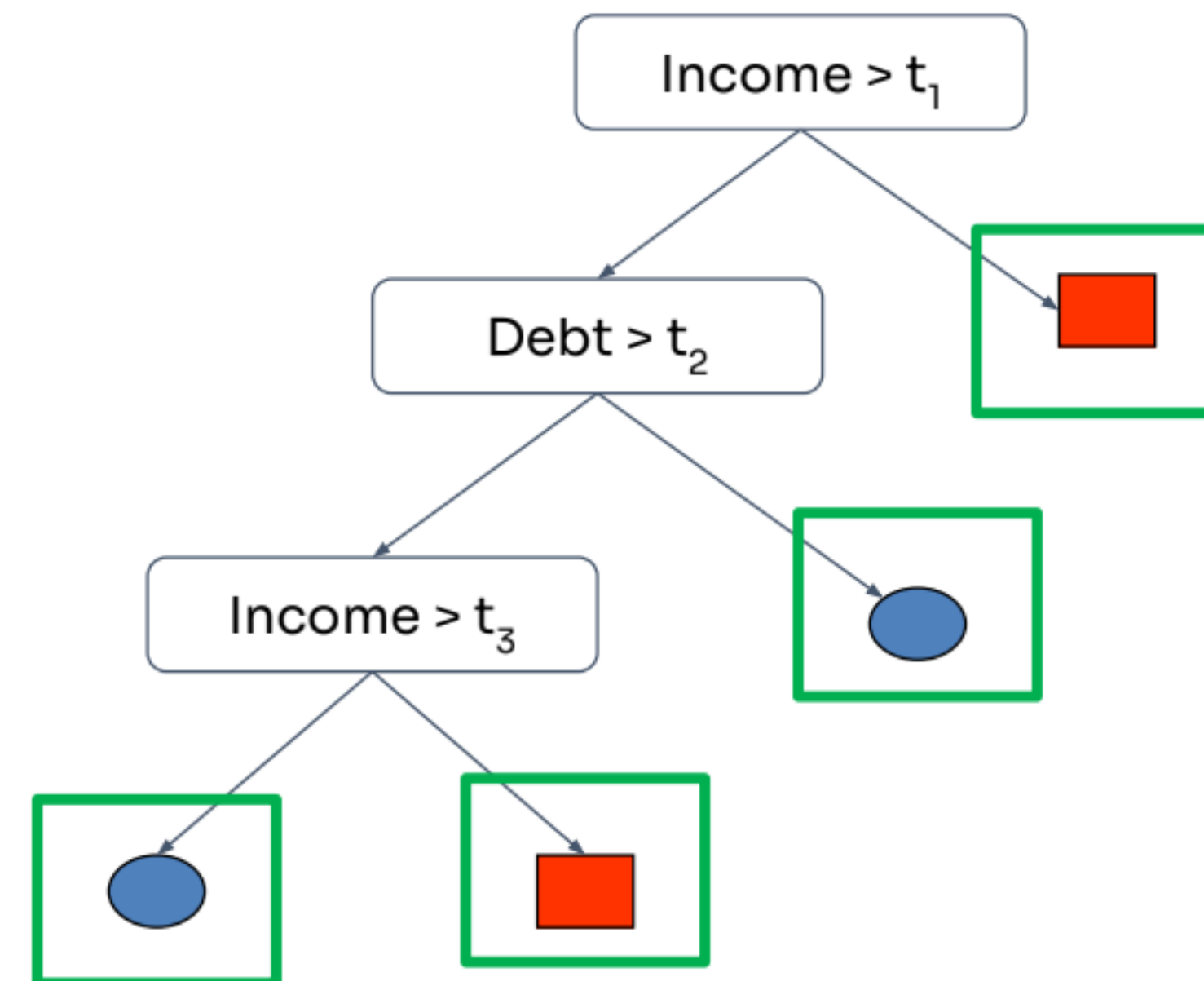
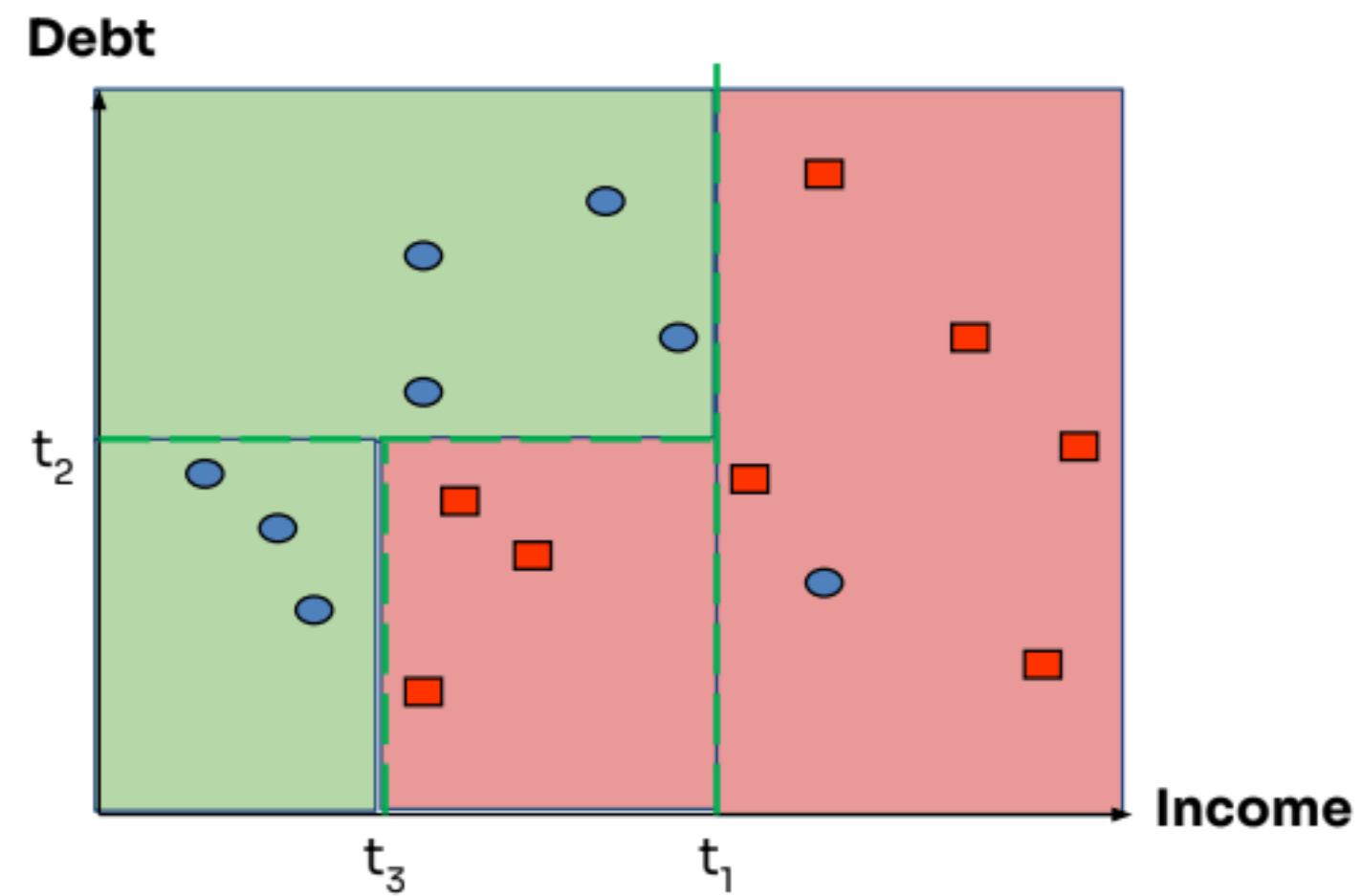
Decision Tree

Decision boundary



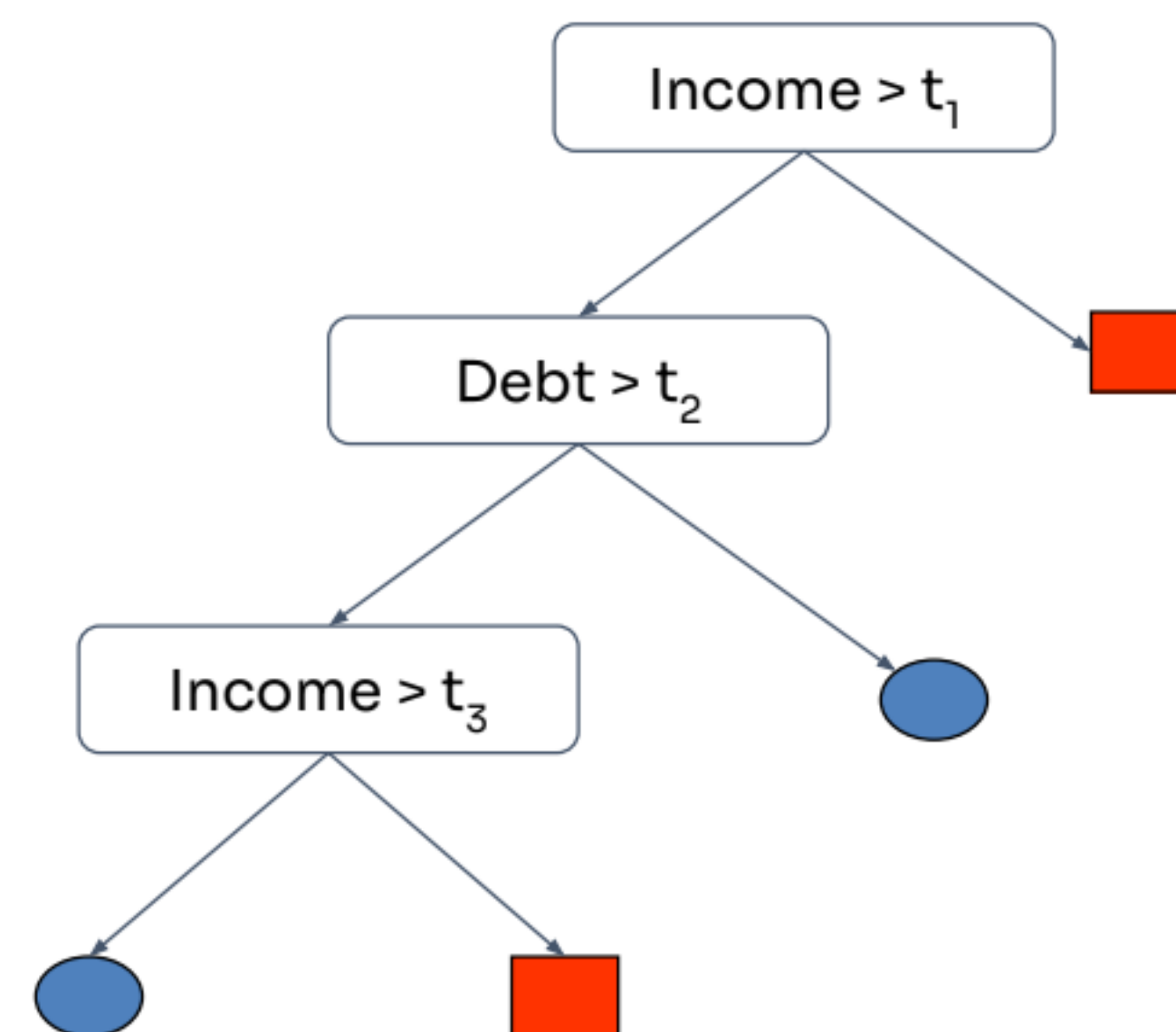
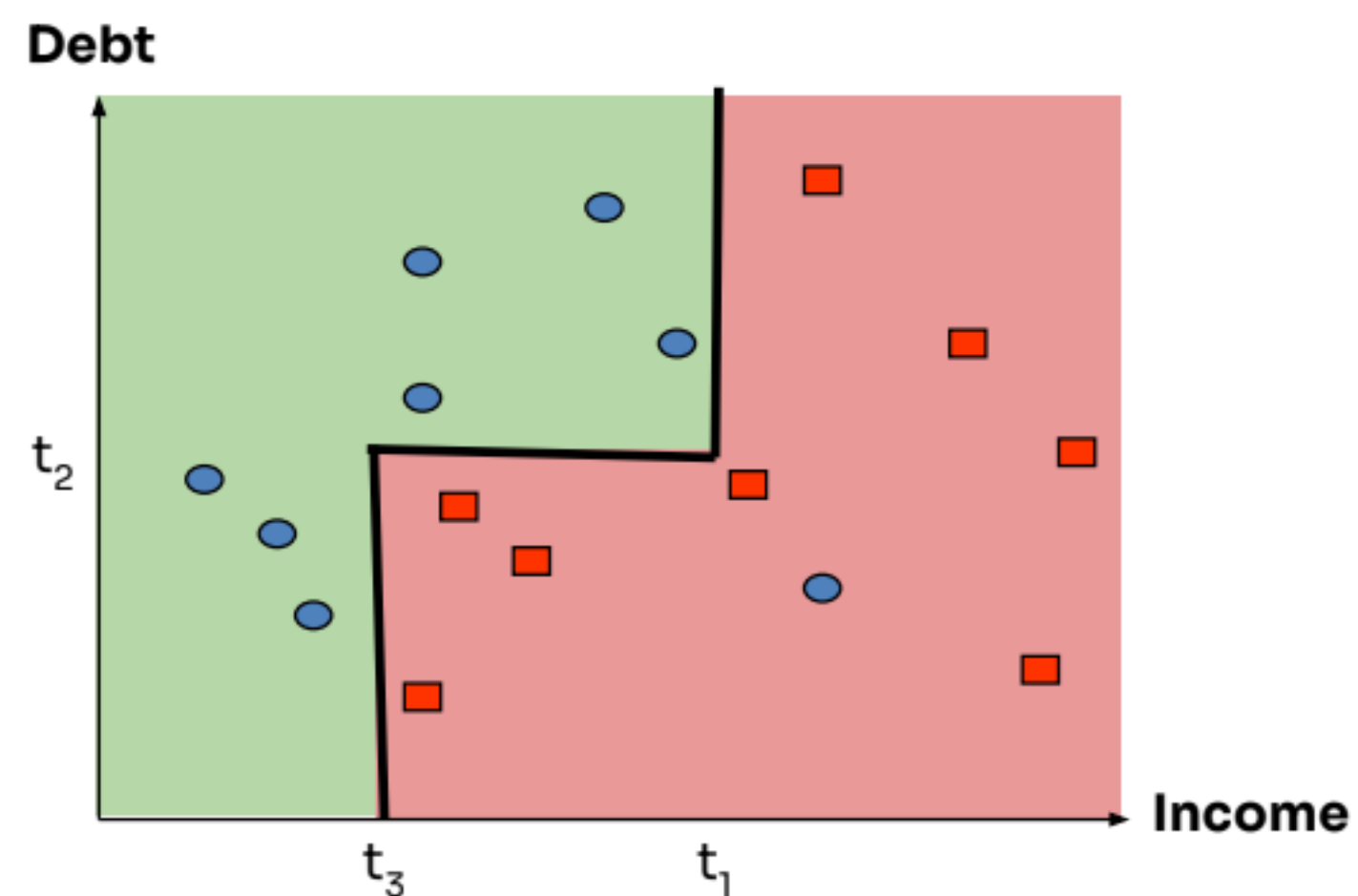
Decision Tree

Decision boundary



Decision Tree

Decision boundary



Decision Tree: Another Example

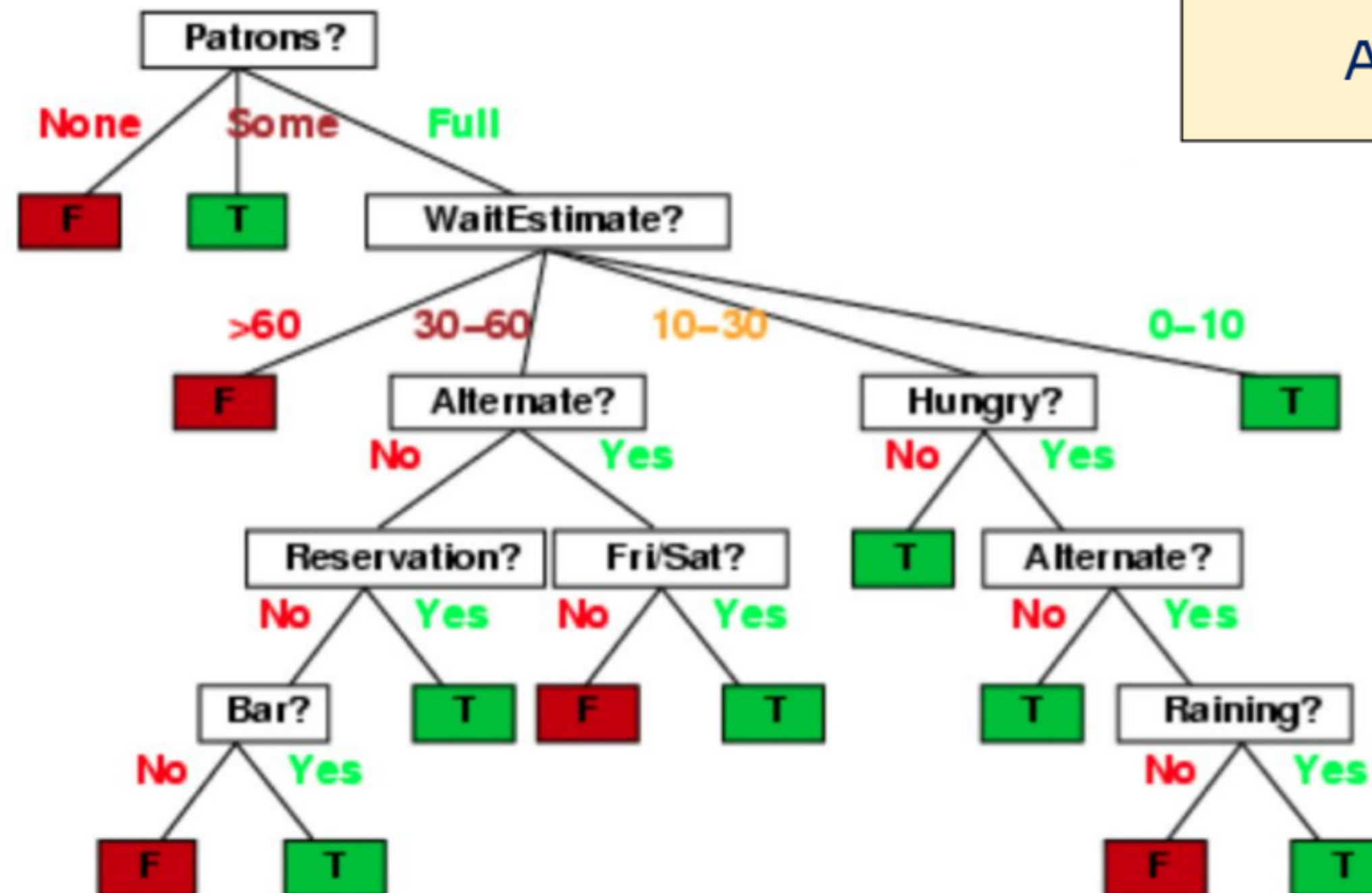
Model a patron's decision of whether to wait for a table at a restaurant

Example	Attributes										Target <i>Wait</i>
	<i>Alt</i>	<i>Bar</i>	<i>Fri</i>	<i>Hun</i>	<i>Pat</i>	<i>Price</i>	<i>Rain</i>	<i>Res</i>	<i>Type</i>	<i>Est</i>	
X_1	T	F	F	T	Some	\$\$\$	F	T	French	0-10	T
X_2	T	F	F	T	Full	\$	F	F	Thai	30-60	F
X_3	F	T	F	F	Some	\$	F	F	Burger	0-10	T
X_4	T	F	T	T	Full	\$	F	F	Thai	10-30	T
X_5	T	F	T	F	Full	\$\$\$	F	T	French	>60	F
X_6	F	T	F	T	Some	\$\$	T	T	Italian	0-10	T
X_7	F	T	F	F	None	\$	T	F	Burger	0-10	F
X_8	F	F	F	T	Some	\$\$	T	T	Thai	0-10	T
X_9	F	T	T	F	Full	\$	T	F	Burger	>60	F
X_{10}	T	T	T	T	Full	\$\$\$	F	T	Italian	10-30	F
X_{11}	F	F	F	F	None	\$	F	F	Thai	0-10	F
X_{12}	T	T	T	T	Full	\$	F	F	Burger	30-60	T

~7,000 possible cases

based on slide by Eric Eaton

Decision Tree: Another Example



Apa ini tree terbaik?

based on slide by Eric Eaton

Choosing the Best Attribute

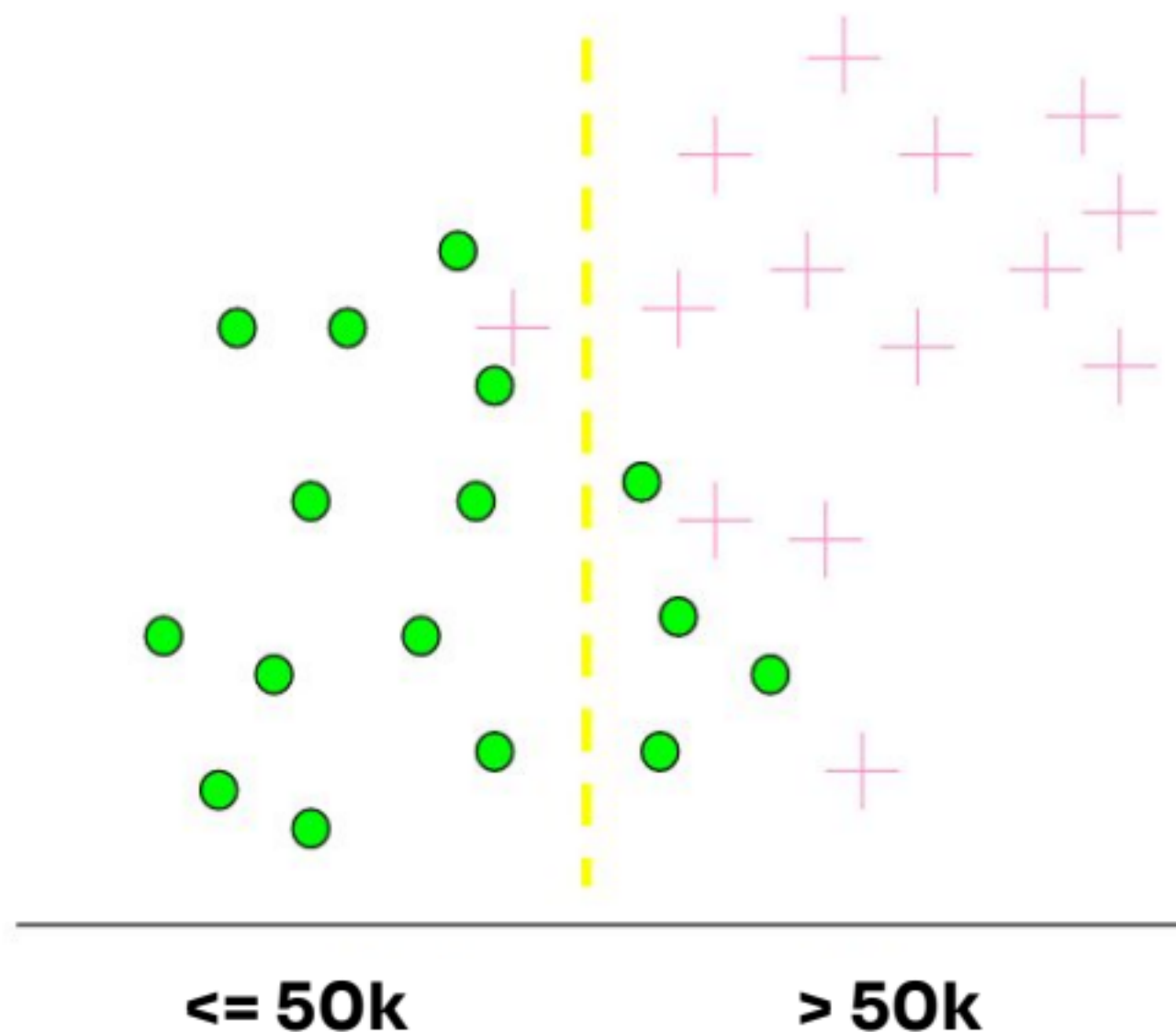
Choosing the best attribute

- **Key problem:** memilih atribut mana yang dipakai untuk memisahkan data.
- Beberapa opsi:
 - **Random** → Pilih atribut secara acak (random)
 - **Least Values** → Pilih atribut dengan jumlah *possible value* terkecil
 - **Most Values** → Pilih atribut dengan jumlah *possible value* terbesar
 - **Max-Gain** → pilih atribut yang memiliki **information gain** terbesar.
 - yang mana akan memperkecil ukuran tree.

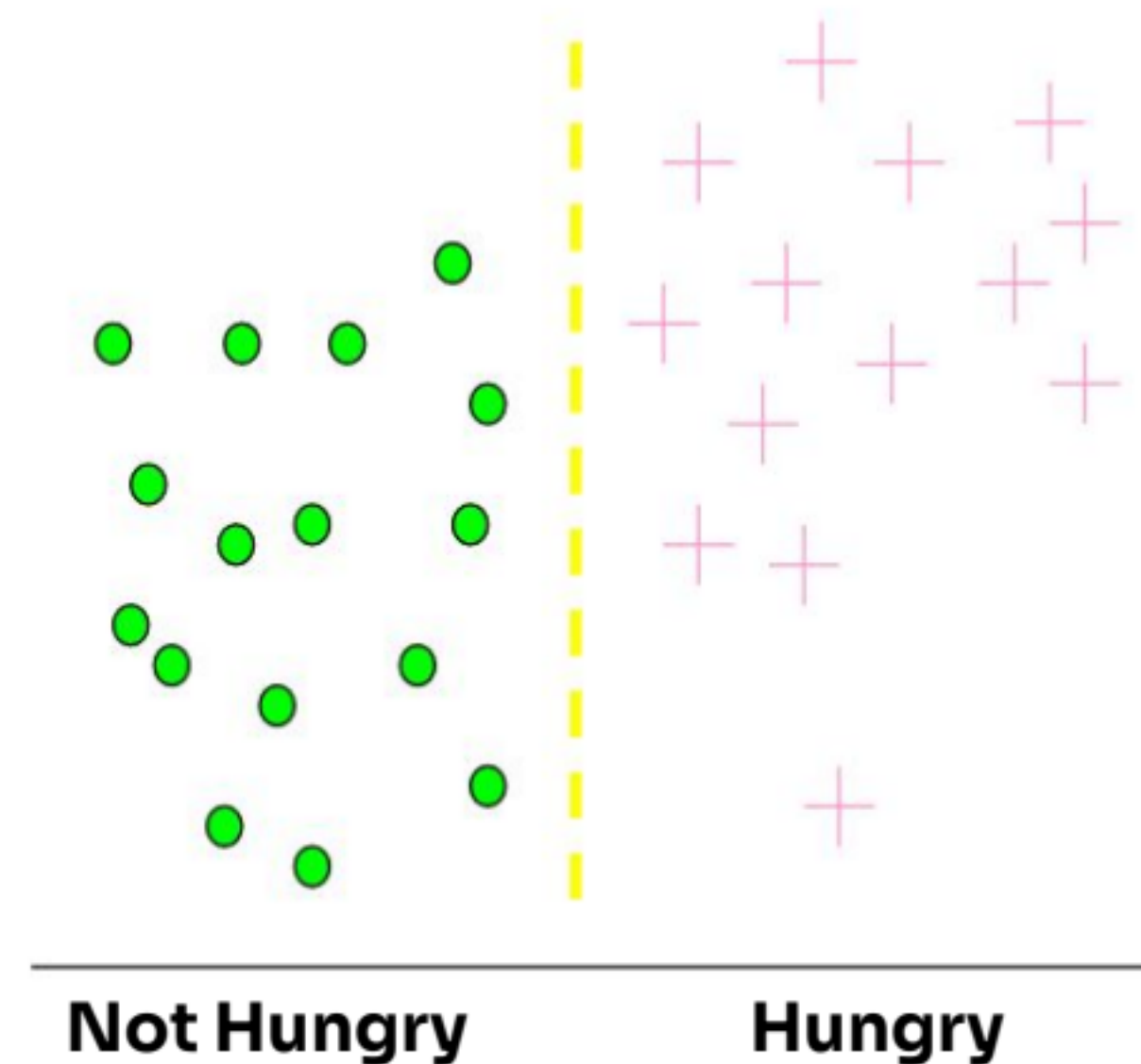
Information Gain

- Uji mana yang lebih informatif?

Membagi apa **Price** lebih dari 50k atau tidak



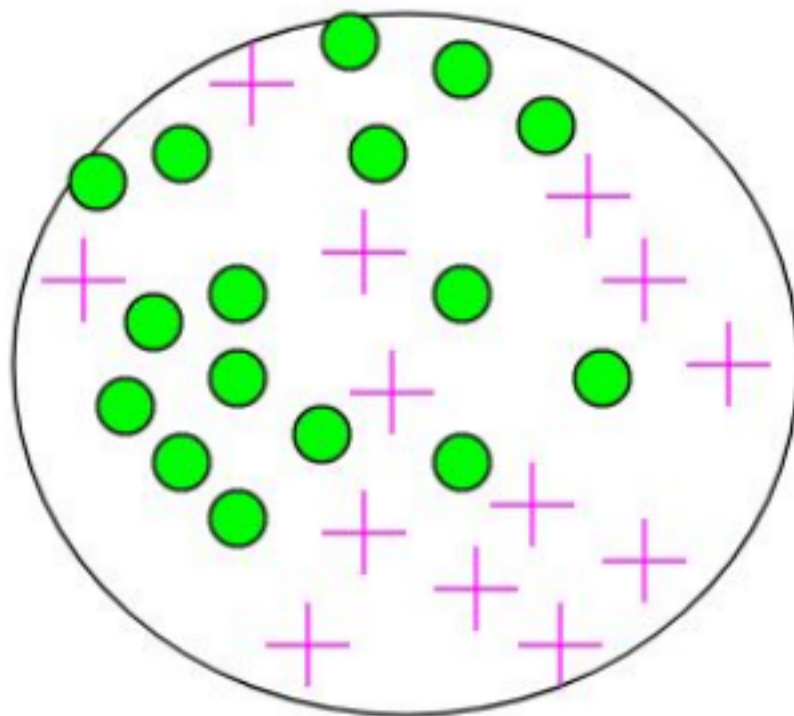
Membagi apa applicant lapar/tidak



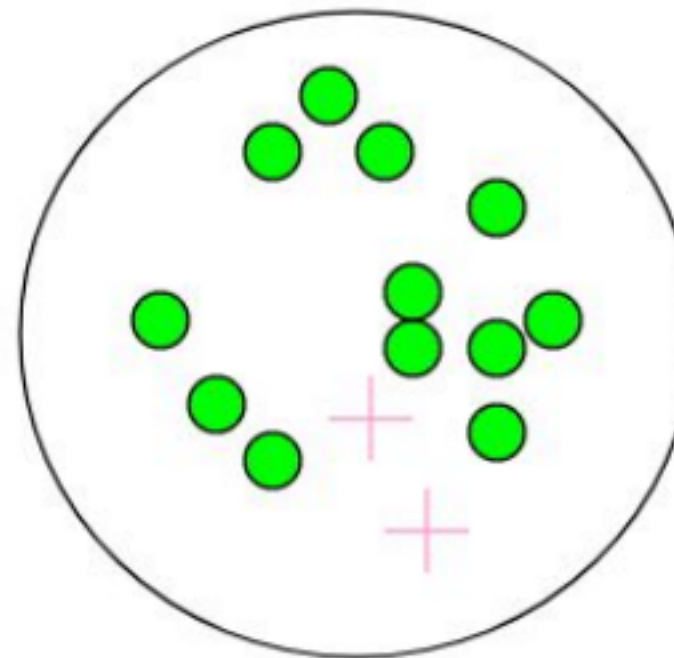
Impurity/Entropy Calculation

- **Entropy** : mengukur tingkat **impurity** dari grup

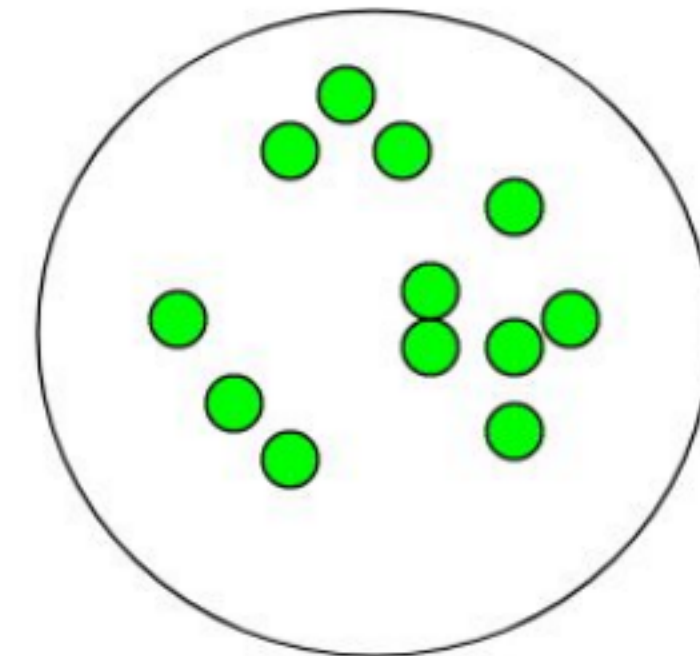
Very impure group



Less impure



Minimum impurity



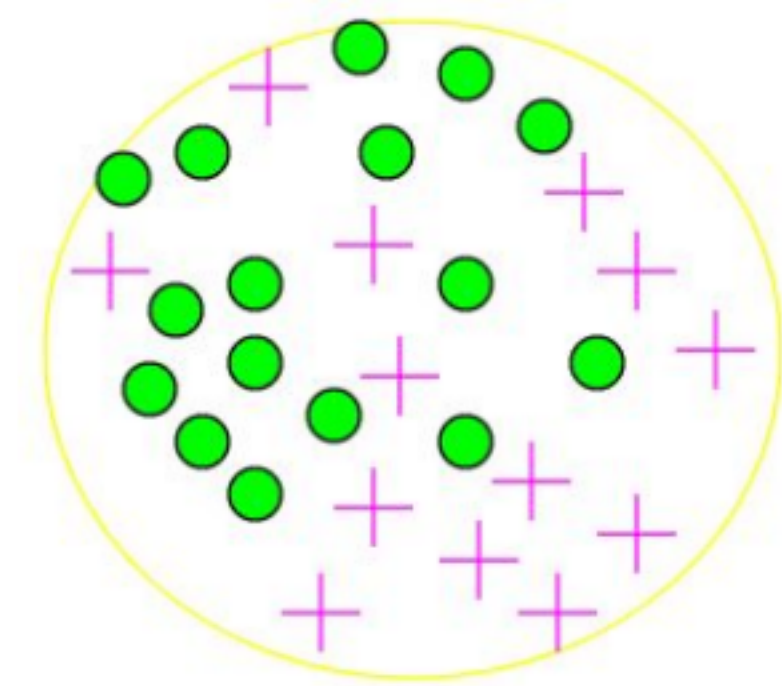
based on slide by Pedro Domingos

Impurity/Entropy Calculation

- Entropy, $H(x)$

$$H(x) = - \sum_{i=1}^n P(x = i) \log_2 P(x = i)$$

$P(x=i)$ adalah probability dari kelas- i . Dihitung sebagai proporsi kelas- i pada set.



Impurity/Entropy Calculation

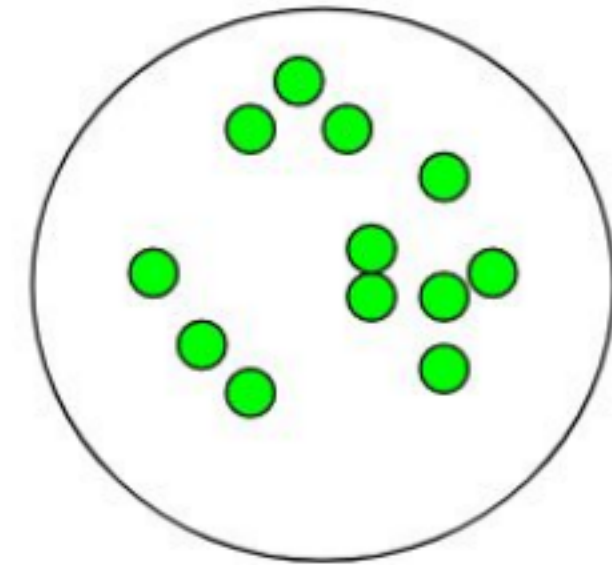
- Berapa entropi dari grup ketika semua anggotanya homogen (ada pada kelas yang sama)?

$$i = 1, P(x = i) = 1.0$$

$$\text{Entropi} = -1 \times \log_2(1) = 0$$

→ MINIMUM IMPURITY

Bukan set yang bagus untuk belajar



- Berapa entropi dari grup ketika anggotanya 50% dari kelas 1 dan sisanya kelas 2?

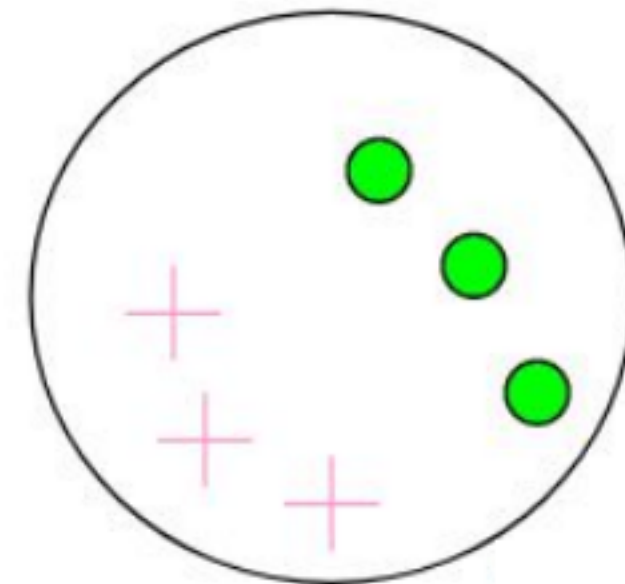
$$i = 1, P(x = i) = 0.5$$

$$i = 2, P(x = i) = 0.5$$

$$\text{Entropi} = -(0.5 \times \log_2(0.5) + 0.5 \times \log_2(0.5)) = 1$$

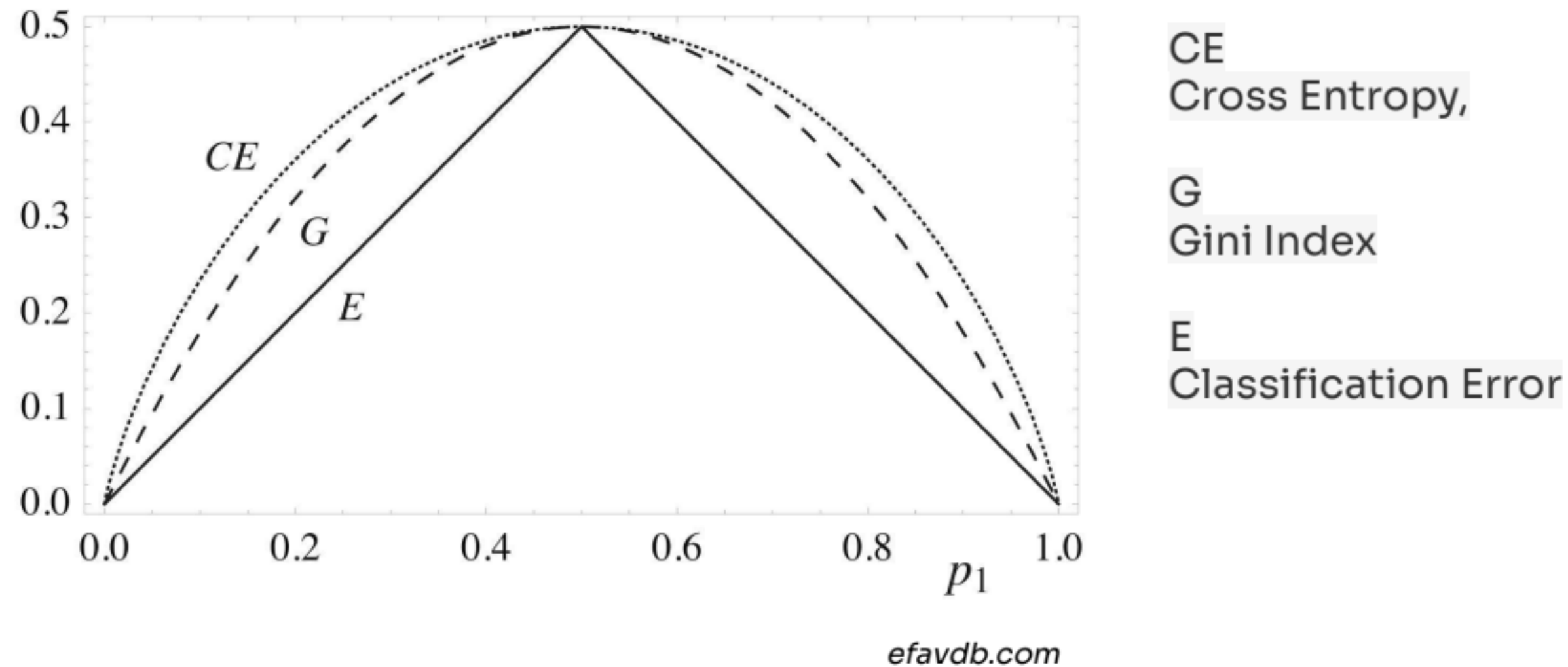
→ MAXIMUM IMPURITY

Set yang bagus untuk decision tree belajar



Measure of Impurity

- Selain entropy, ada juga gini index & classification error untuk menghitung error



Decision Tree: Another Construction

Model a patron's decision of whether to wait for a table at a restaurant

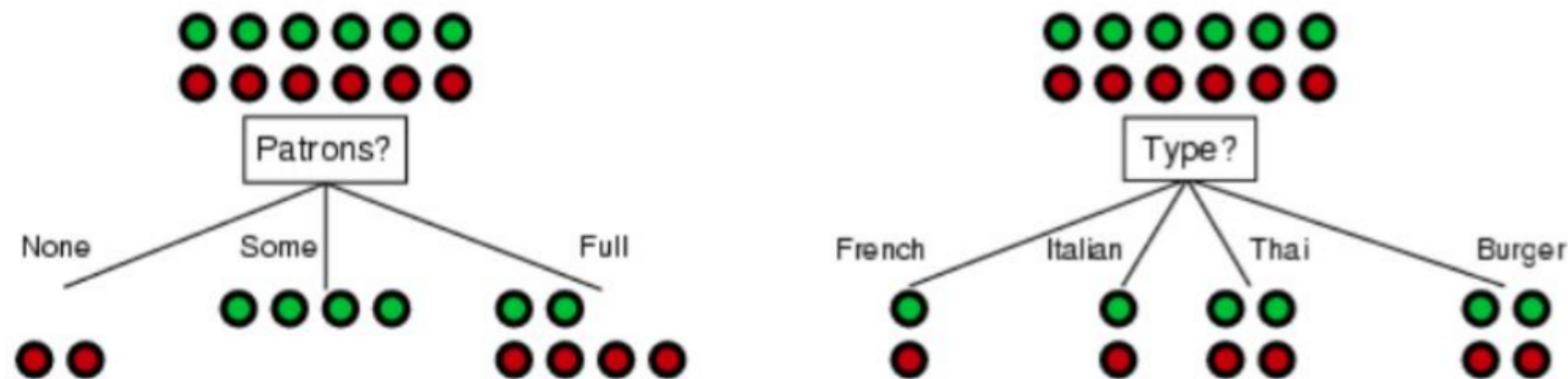
Example	Attributes										Target <i>Wait</i>
	<i>Alt</i>	<i>Bar</i>	<i>Fri</i>	<i>Hun</i>	<i>Pat</i>	<i>Price</i>	<i>Rain</i>	<i>Res</i>	<i>Type</i>	<i>Est</i>	
X_1	T	F	F	T	Some	\$\$\$	F	T	French	0-10	T
X_2	T	F	F	T	Full	\$	F	F	Thai	30-60	F
X_3	F	T	F	F	Some	\$	F	F	Burger	0-10	T
X_4	T	F	T	T	Full	\$	F	F	Thai	10-30	T
X_5	T	F	T	F	Full	\$\$\$	F	T	French	>60	F
X_6	F	T	F	T	Some	\$\$	T	T	Italian	0-10	T
X_7	F	T	F	F	None	\$	T	F	Burger	0-10	F
X_8	F	F	F	T	Some	\$\$	T	T	Thai	0-10	T
X_9	F	T	T	F	Full	\$	T	F	Burger	>60	F
X_{10}	T	T	T	T	Full	\$\$\$	F	T	Italian	10-30	F
X_{11}	F	F	F	F	None	\$	F	F	Thai	0-10	F
X_{12}	T	T	T	T	Full	\$	F	F	Burger	30-60	T

~7,000 possible cases

based on slide by Eric Eaton

Choosing an attribute

- **Ide**, atribut yang baik membagi data menjadi subset yang ideal, “semua positif” atau “semua negatif”



Based on slide by Pedro Domingos

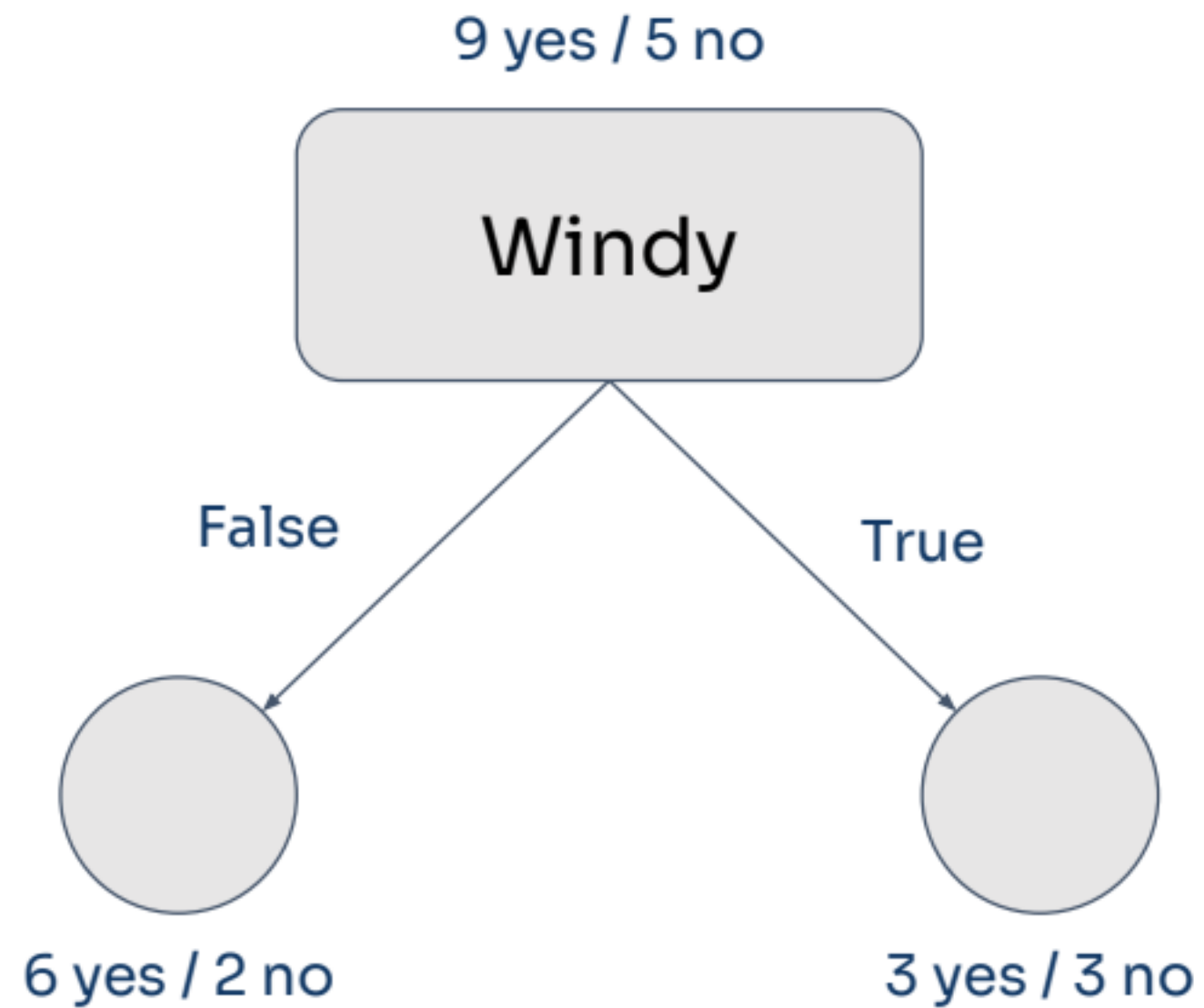
- Pembagian mana yang lebih **informatif?** Patrons atau Type?

Choosing **next** attribute

- **Ide**, cari atribut selanjutnya yang paling menurunkan entropi diketahui atribut sebelumnya.
- Ini disebut dengan **information gain**

information gain = **entropy**(parent) – [**average entropy** (children)]

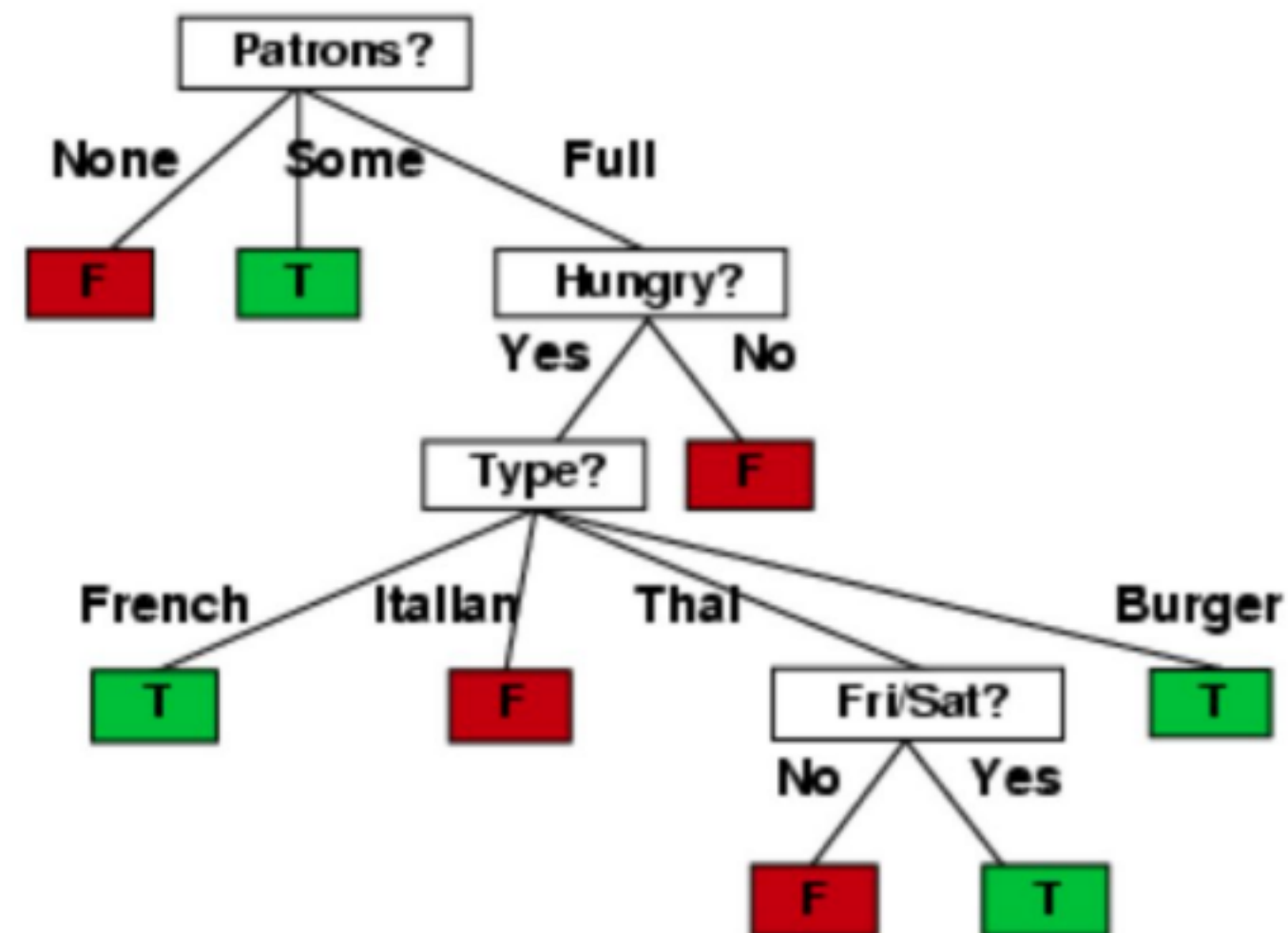
Choosing **next** attribute



$$H(S) = 0.94, H(S_{False}) = 0.81, H(S_{True}) = 1.0$$

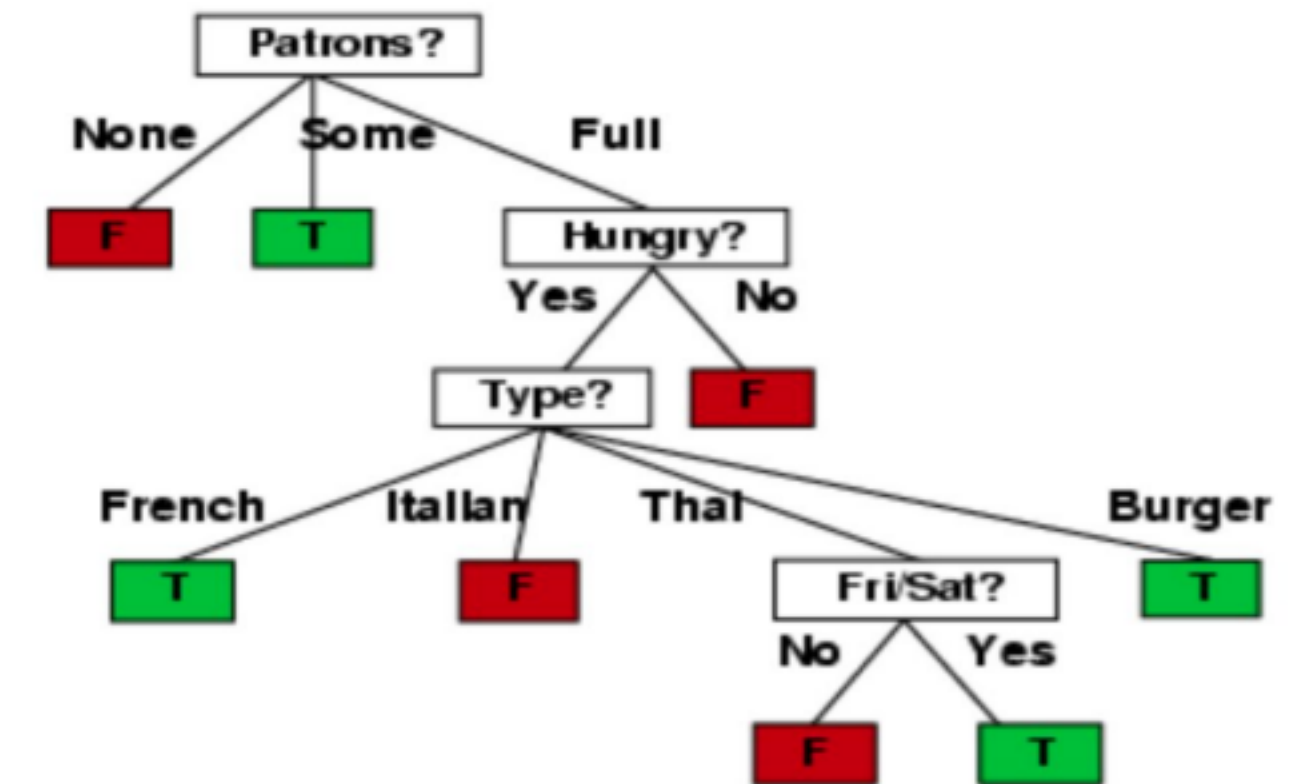
$$Gain(S, Windy) = 0.94 - \frac{8}{14}0.81 - \frac{6}{14}1.0 = 0.049$$

After choosing the best attribute



based on slide by M. desJardins & T. Finin

Compare the two decision tree

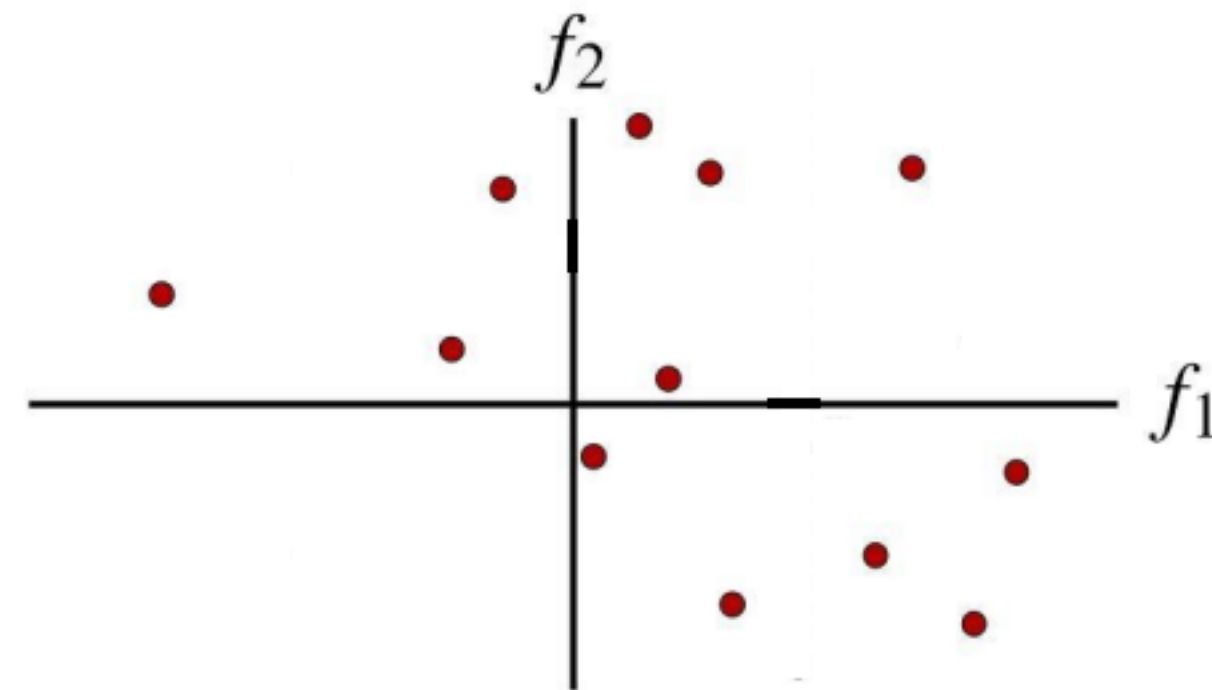


based on slide by M. desJardins & T. Finin

Regression Trees

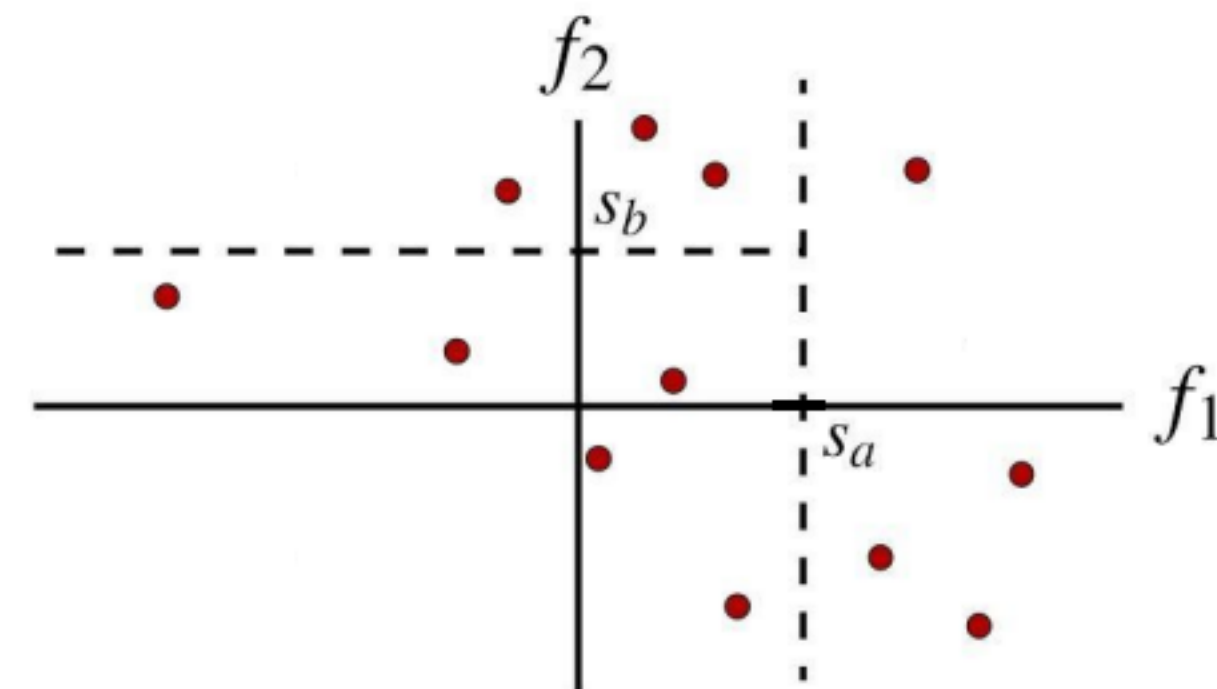
Regression Tree

- Kita punya 2 variable/fitur input, f_1 dan f_2 , untuk prediksi tingkat pendapatan (income)



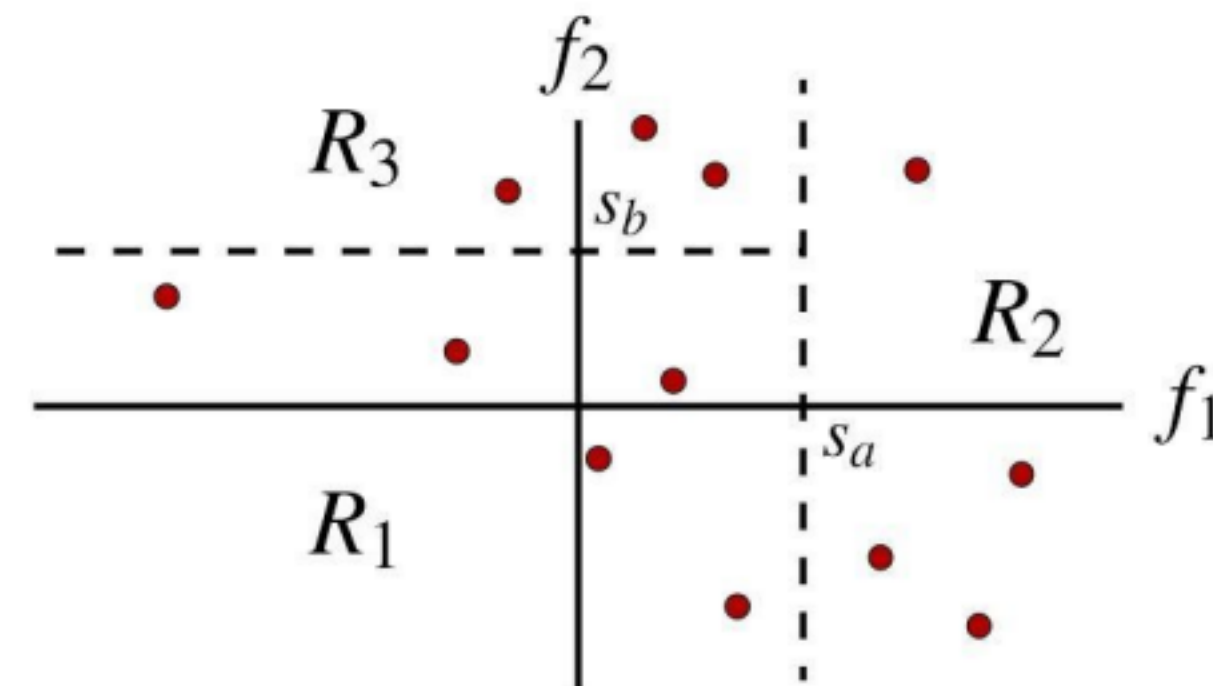
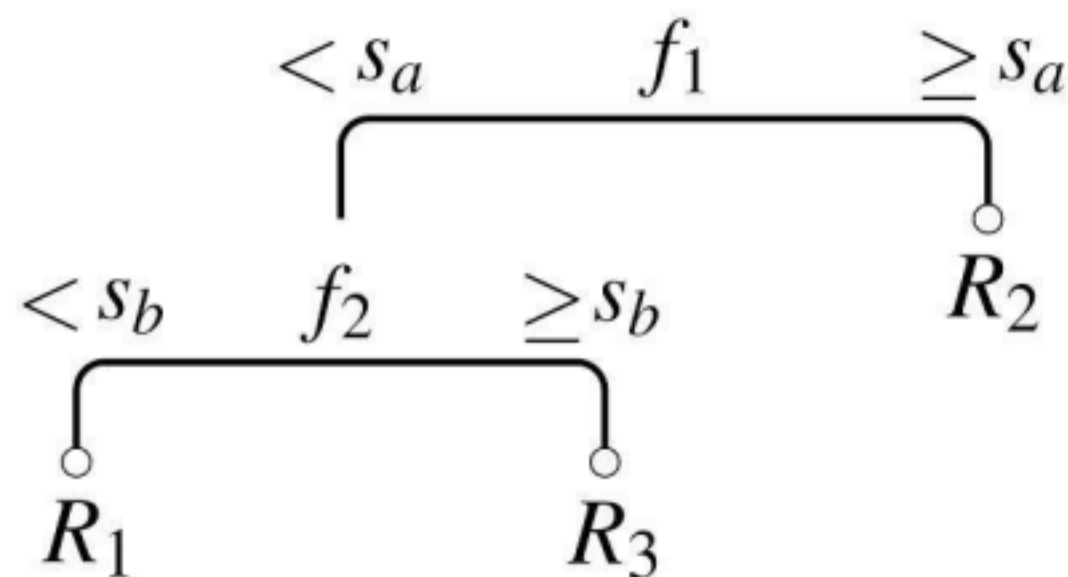
Regression Tree

- Kita punya 2 variable/fitur input, f_1 dan f_2 , untuk prediksi tingkat pendapatan (income)
- Pembagi pertama adalah s_a pada f_1 dan pembagi kedua adalah s_b pada f_2



Regression Tree

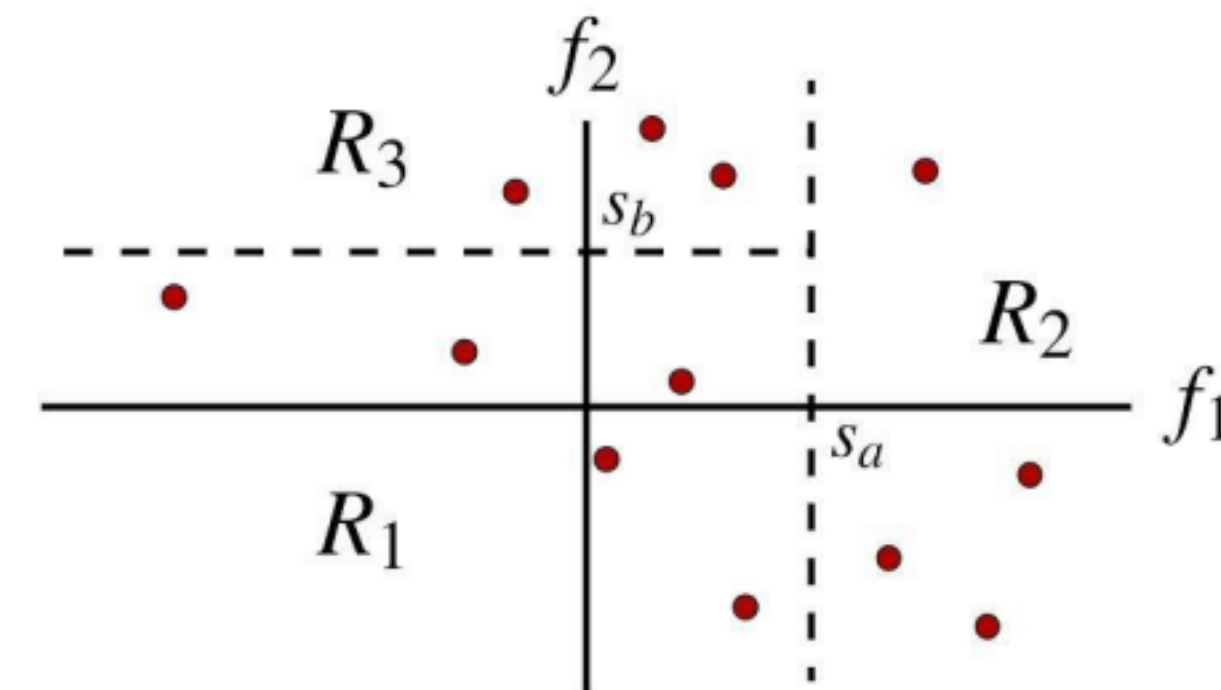
- Kita punya 2 variable/fitur input, f_1 dan f_2 , untuk prediksi tingkat pendapatan (income)
- Pembagi pertama adalah s_a pada f_1 dan pembagi kedua adalah s_b pada f_2
- Tree akan membagi feature space menjadi 3 region, R_1 , R_2 , dan R_3



Regression Tree

- Untuk setiap region, prediksi kita adalah **rata-rata (mean) dari data target di masing-masing region**.
- Ingat **K-Nearest Neighbor Regression**?
- Dari observasi berikut, income level adalah rata-rata data di region R_i

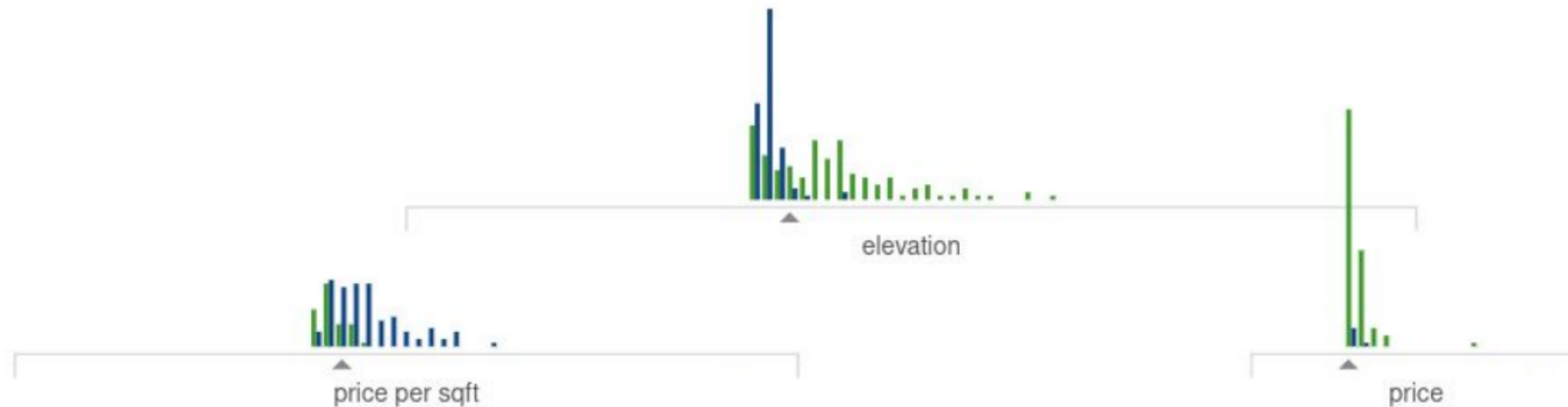
Region	Mean of Training Label
R_1	Rp. 4.560.000
R_2	Rp. 6.840.000
R_3	Rp. 10.940.000



Overfitting

Growing Tree

- Penambahan branch/tree akan **meningkatkan akurasi** prediksi.



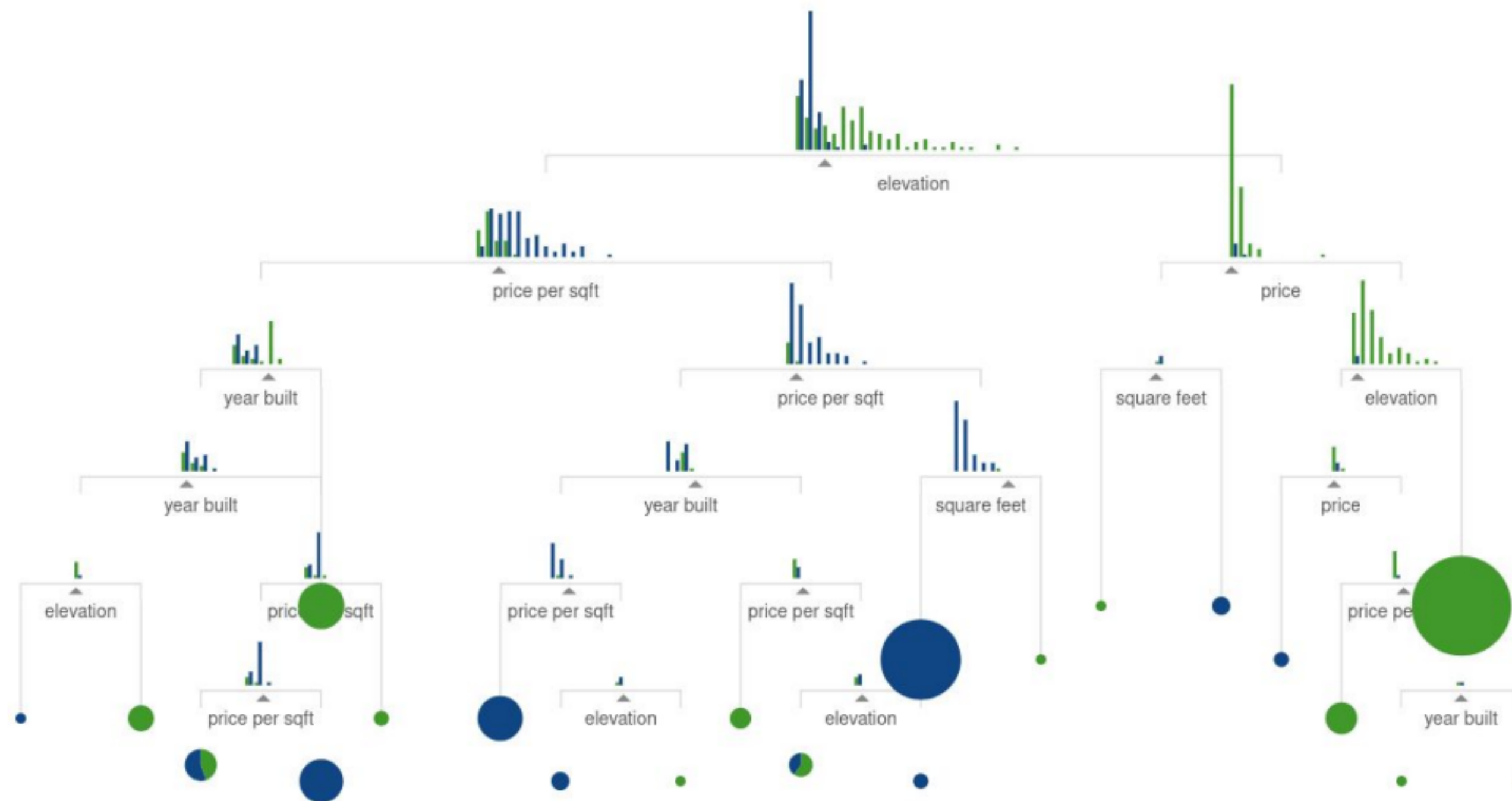
Tree Depth: 3

- Split 1 layer lebih dalam, akurasi tree meningkat menjadi **84%**



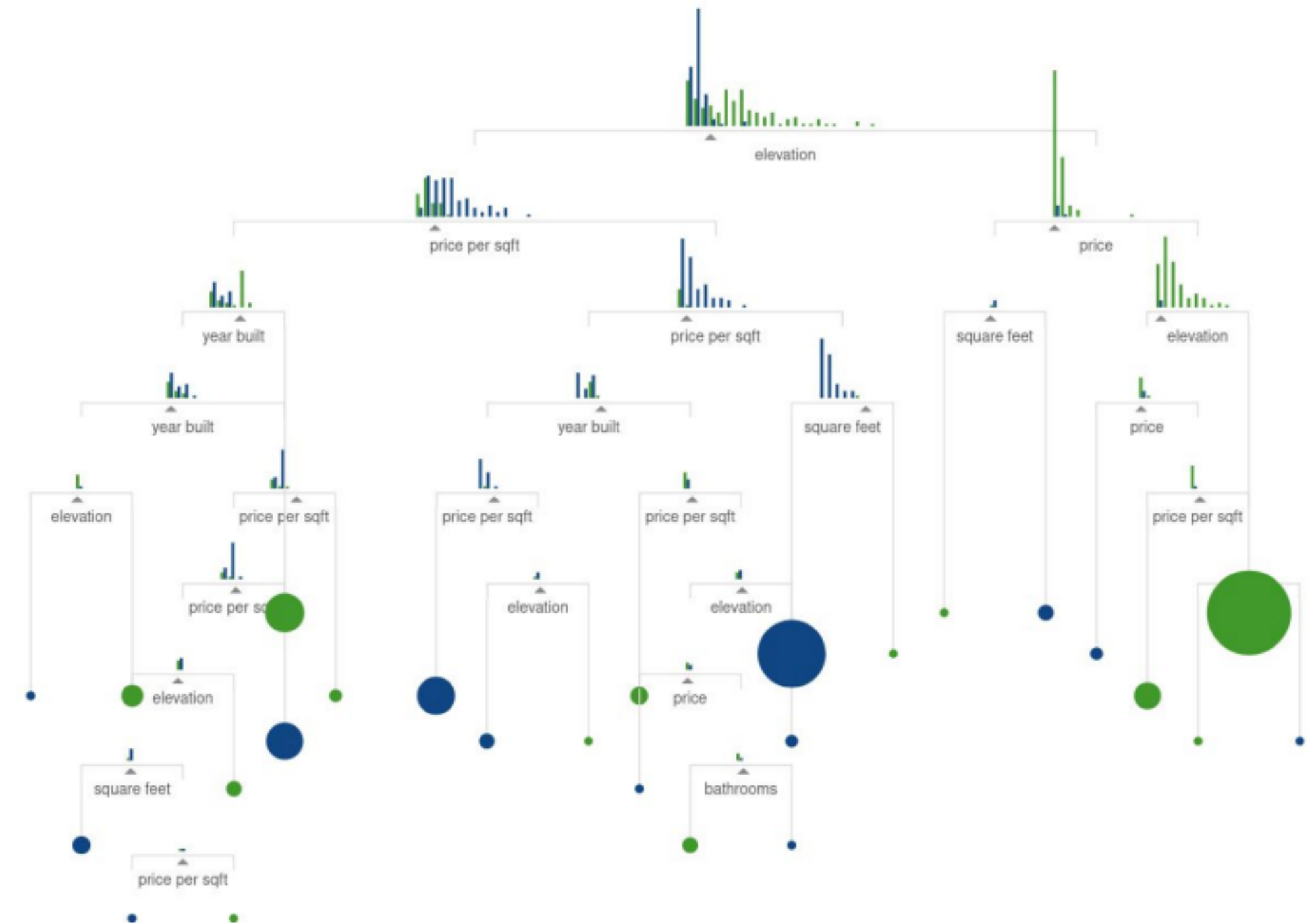
Tree Depth: 6

- Menambahkan beberapa layer lagi, kita dapatkan akurasi **96%**

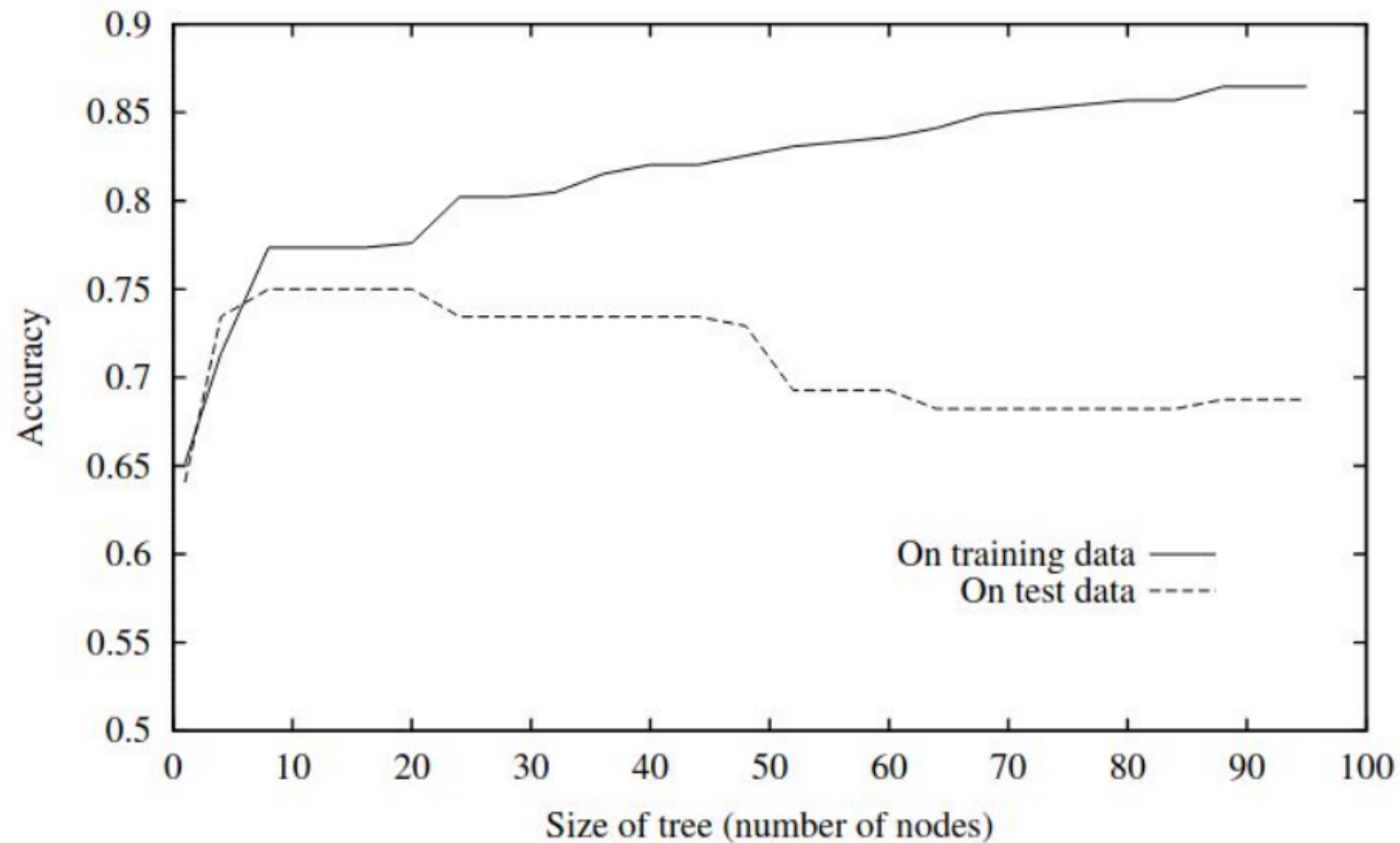


Overfitting

- Apabila dilanjutkan, akurasi akan **100%**
- Akurasi ini terjadi di data training.
- Namun ternyata, akurasi di data testing (future data) kecil.



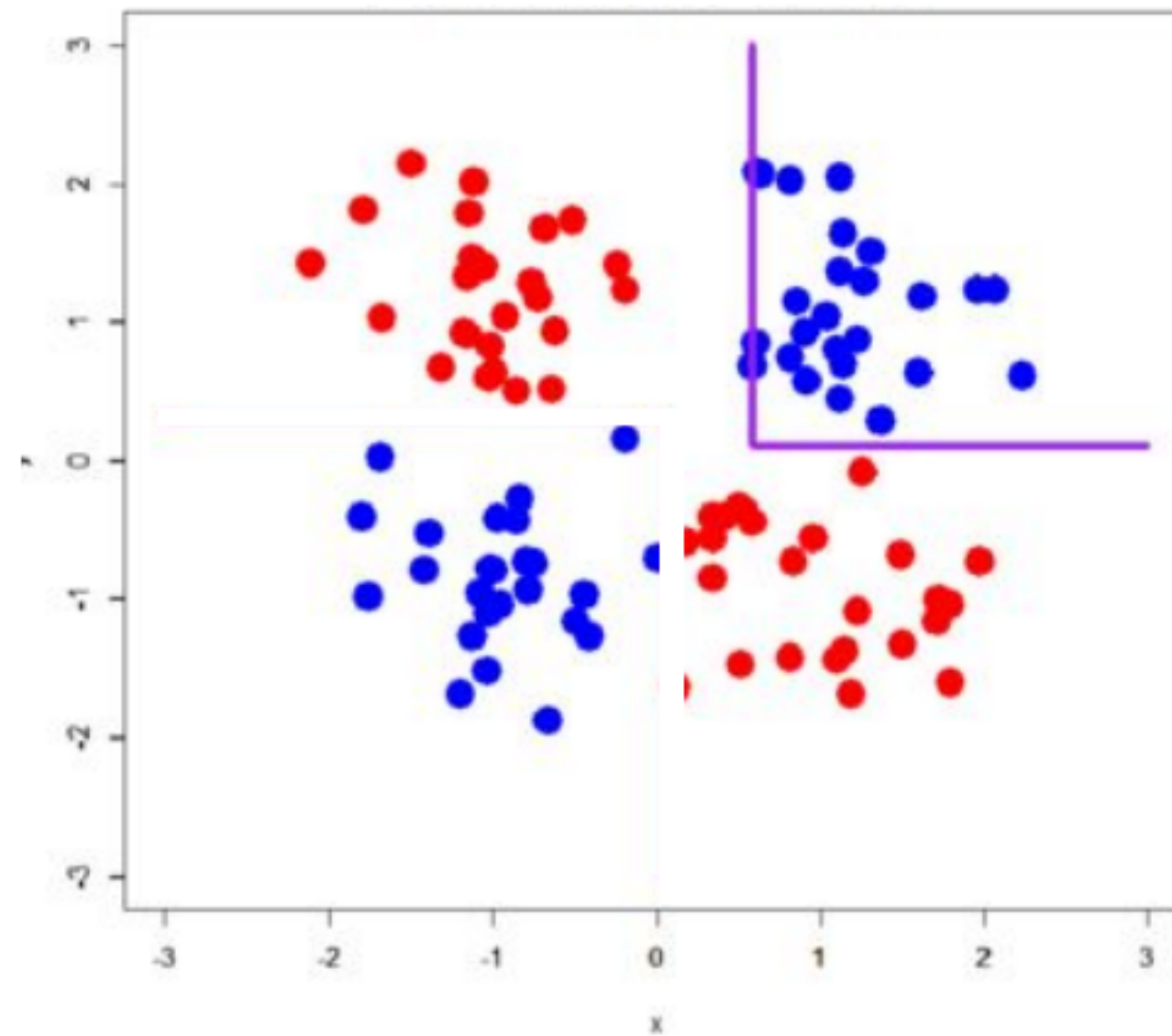
Overfitting



Tree Complexity

Simple Tree

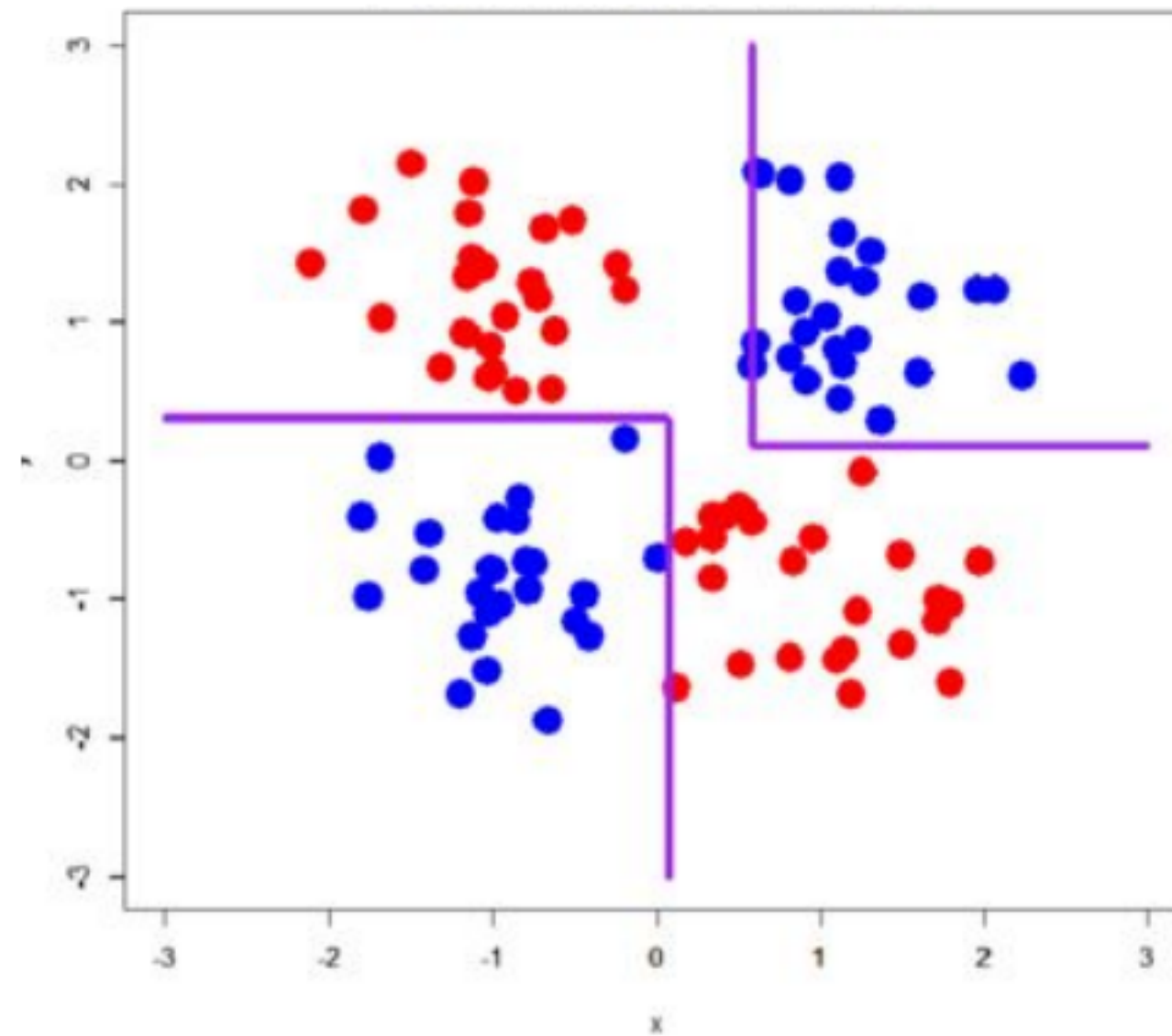
- Decision boundary yang sederhana



Takashi J. Ozaki, Ph. D.

Less Simple Tree

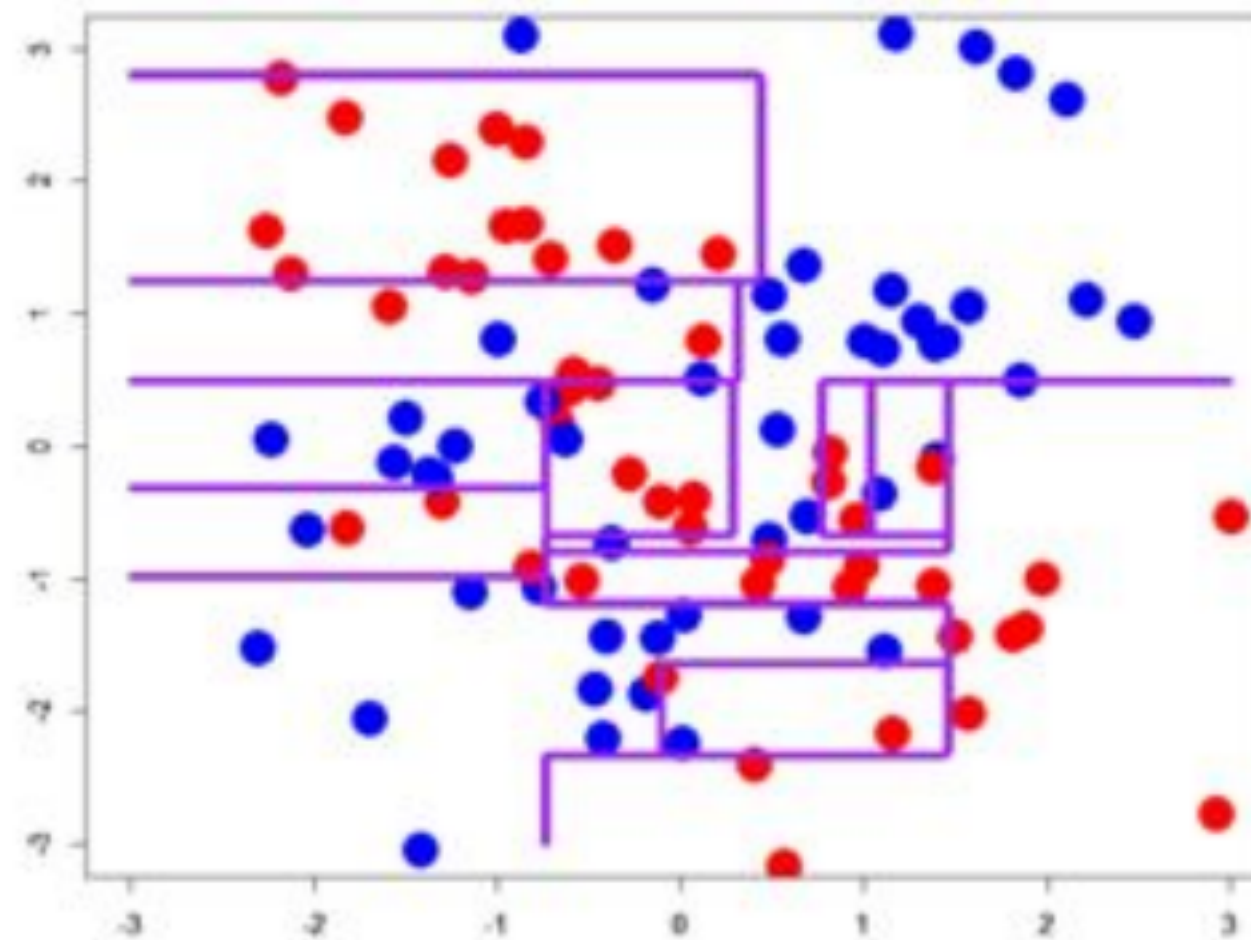
- Decision boundary yang sedikit lebih kompleks



Takashi J. Ozaki, Ph. D.

Less Simple Tree

- Decision boundary yang sangat kompleks



Takashi J. Ozaki, Ph. D.

Tree Pruning

Overfitting

- Decision tree sangat rentan untuk **overfitting**
- Tanpa **stopping criterion** selama proses pembuatan tree, Anda akan mudah mendapatkan **akurasi 100%** pada data training.
- Bagaimana cara mengatasi ini?

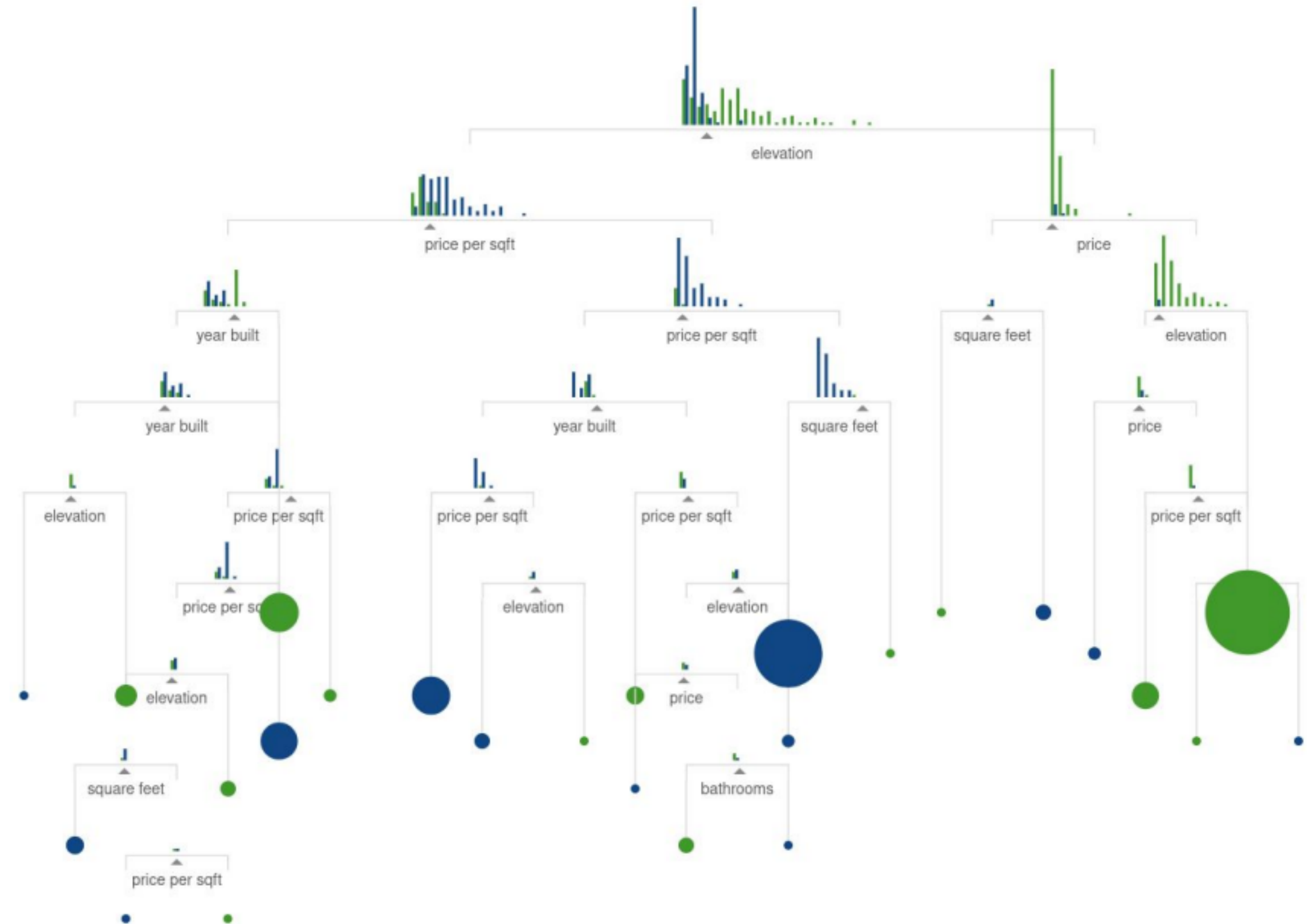
Overfitting



Potong branch-nya

How to prune?

- **Pertama**, tumbuhkan tree “super-complex”



How to prune?

- **Kedua**, pilih tuning parameter alpha (non-negative float)
- Untuk tiap nilai dari alpha, akan ada subset T sedemikian sehingga

$$\sum_{m=1}^{|T|} \sum_{i: x_i \in R_m} (y_i - \hat{y}_{R_m}) + \alpha |T|$$

bernilai semimumimum mungkin



Stopping criteria

- Pruning dari performa **validation set** – prune subtree **untuk** memaksimalkan performa di validation set

Limitation of Trees

Why trees are widely used in practice

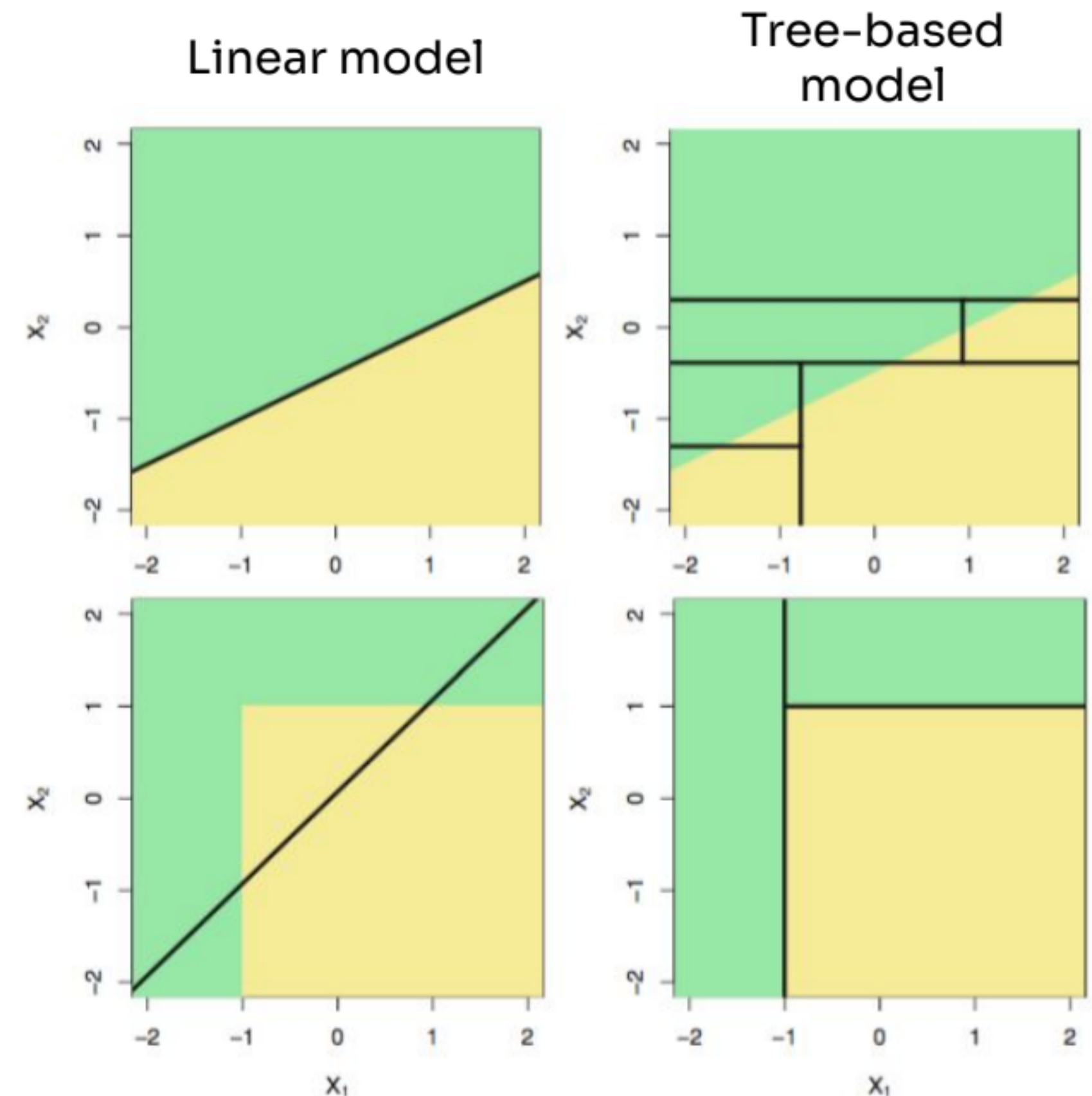
- Dapat memodelkan **high dimensional data**
- Trees dapat diinterpretasikan
 - Domain expert dapat **membaca** logika tree sebagai “rules”
- Algoritma tree mudah untuk di **code** dan diuji

Limitation of trees

- Kesulitan dalam memodelkan struktur linear
- Kurang *smooth*

Linear boundary

Non-linear boundary



Limitation of trees

- Variasi model yang tinggi
 - Trees tergolong “tidak stabil” sebagai fungsi dari sampel
 - Akibatnya:
 - Variasi besar (high variance) pada prediksi error.
 - Variasi besar yang menurunkan interpretabilitas.
 - Trees dapat dijadikan kandidat bagus untuk gabungan model
 - sering digunakan dengan boosting & bagging.

Decision trees are not stable

- Merubah satu data “sedikit” dapat mengakibatkan perbedaan yang cukup signifikan terhadap partisi.

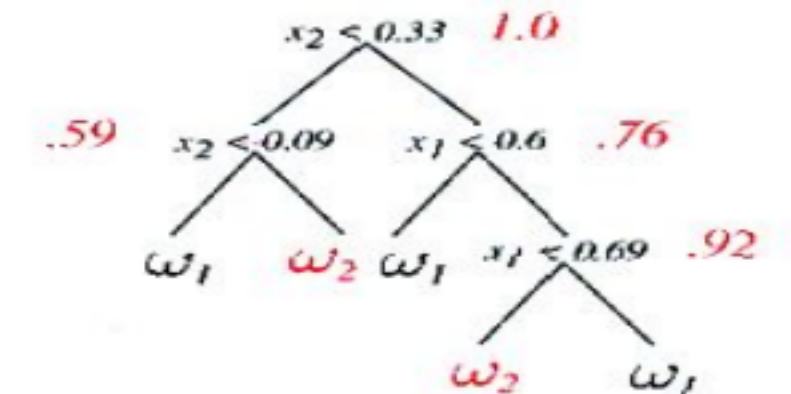
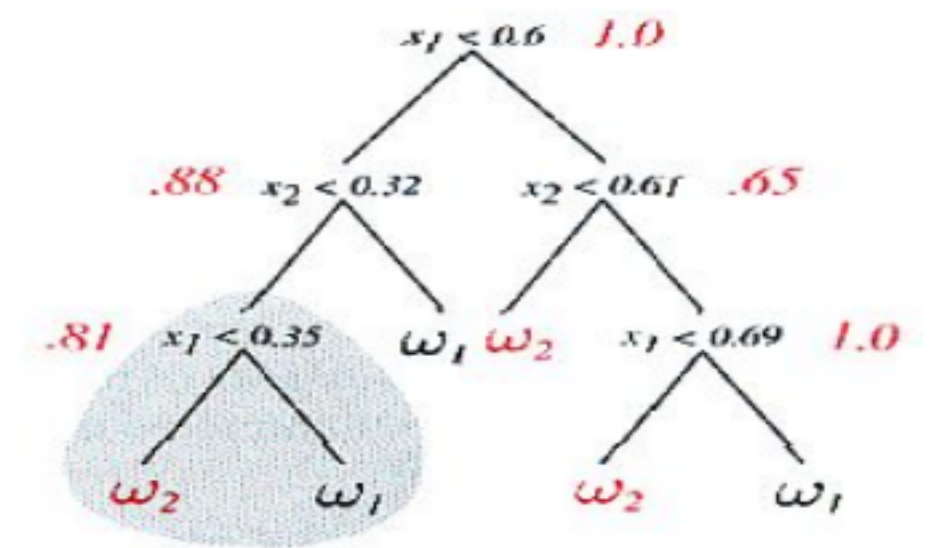
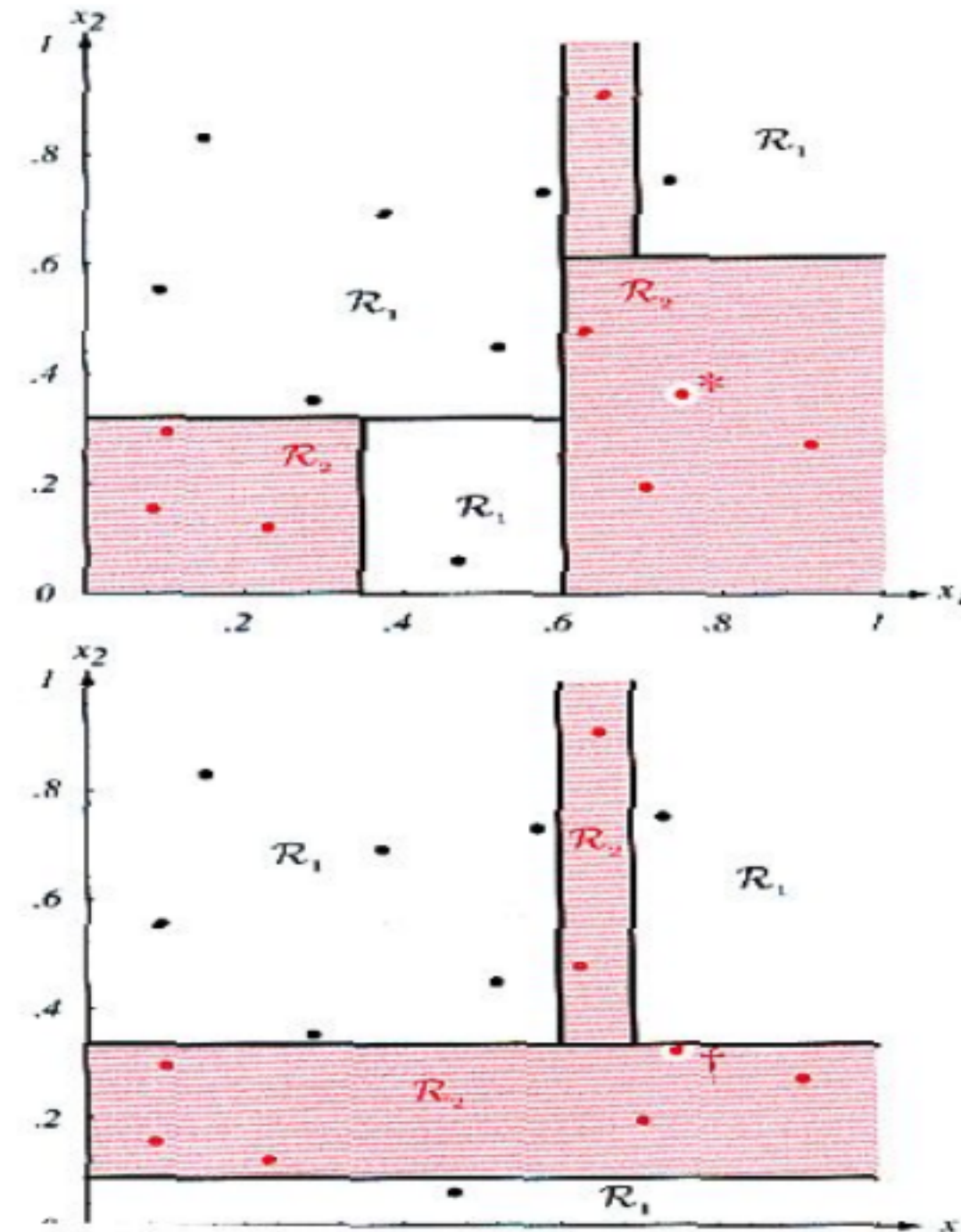


Figure from Duda, Hart & Stork, Chap. 8
Data mining- Volinsky - 2011 - Columbia University

Thank You
